



BITS Pilani
Pilani Campus

Microwave Engineering (ECE F331)

Prof. Pankaj Arora
Department of Electrical and Electronics Engineering



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Pilani Campus

Electromagnetic Fields and Microwave Engineering (ECE F314)

Prof. Pankaj Arora
Department of Electrical and Electronics Engineering

Course Handout



Course Description:

Course No.

Course Title

Instructor-in-charge

: ECE F314 or ECE F331

**: Electromagnetic Fields and
Microwave Engineering**

: Prof. Pankaj Arora

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(Chamber: 2210-C, EEE Dept.)

: Prof. Dinesh Rano

Instructors (For Tutorials) :

sec-1: Prof. Dinesh Rano : Room 6102

sec-2: Prof. Pankaj Arora : Room 1202

Text Books:

- 1) TB1: David Pozar, “Microwave Engineering”, 4th edition, John Wiley & Sons, 2012.
- 2) TB2: John D. Kraus and Daniel A. Fleisch, “Electromagnetics”, 5th ed., McGraw-Hill, New York, 1999.

Reference Books:

- i) RB1: David K. Cheng, “Field and Wave Electromagnetics” 2nd ed. Pearson Education, New Delhi, 2009.
- ii) RB2: Samuel Y. Liao, “Microwave devices and circuits,” 3rd ed. , PHI 2008.
- iii) RB3: J. D. Krauss et al., “Antennas and Wave Propagation”, 4th edition, TMH 2010.

Evaluation Scheme (ECE F314):

Component	Duration	Marks (200)	Weightage	Evaluation type
Mid-term Test	90 min	60	30%	Closed Book
Quizzes	TBA	60	30%	Closed Book
Comprehensive Examination	3 hours	80	40%	OB + CB

Evaluation Scheme (ECE F331):

Component	Duration	Marks (200)	Weightage	Evaluation type
Mid-term Test	90 min	60	30%	Closed Book
Quizzes	TBA	30	15%	Closed Book
Lab	TBA	30	15%	Close/Open book
Comprehensive Examination	3 hours	80	40%	OB + CB

Why this course?



- Electrical Science, Microelectronic circuits, ADVD etc. (Amplifiers/Oscillators): Low frequency signal
- What about high frequency signal? Consequences ?

→ High frequency signals travel long distances

- For wireless communication: less height of antennas
- To destroy enemies' drones, radars etc. using high frequency signal
- Heating using high frequency signals e.g. Microwave, Oven
- Fusion heating to provide kinetic energy with high frequency signal
- Cargos scanners with high frequency signals
- Medical technologies: for cancer tumors
- Crowd Controls with high frequency signals

What we are going to learn in this course?



- How to generate high frequency signals? {Oscillators??}
- Or
- How to generate high frequency-high power signals?
- How to transmit high frequency signals? {Copper wires??}
- How to radiate high frequency signals? {Antenna??}

Course Content:



- Transmission Lines and Graphical Analysis
- EM Waves and Waveguides
- Microwave Generation and Amplification
- Antennas

Transmitting the signal from Point A to B (Transmission Lines)

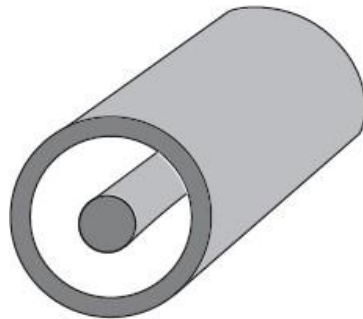
e.g.: Power lines (Copper wires-50 Hz), Telephone Lines(Twisted paired wires-8 to 20kHz), TV Cables (Co-axial wires-MHz), Internet lines (Optical fibers-THz)

Main Difference: Frequency of operation
{Change in Medium}

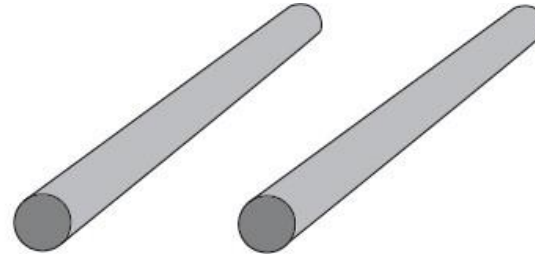
Examples of Transmission Lines:



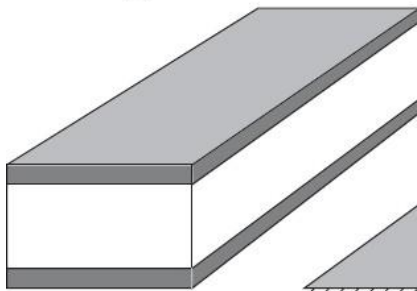
Cross-sectional view



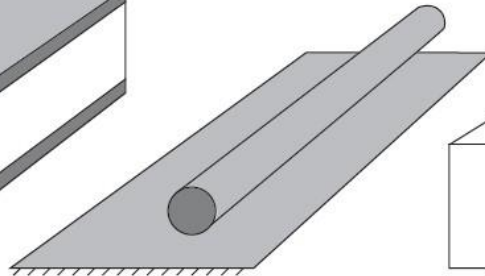
(a)



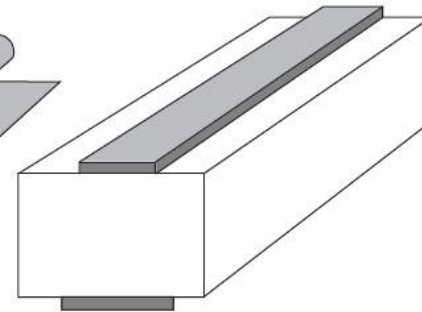
(b)



(c)



(d)



(e)



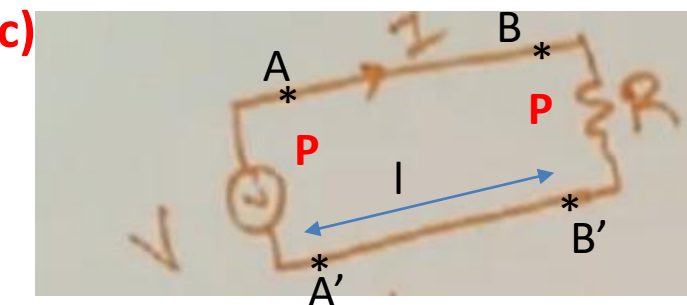
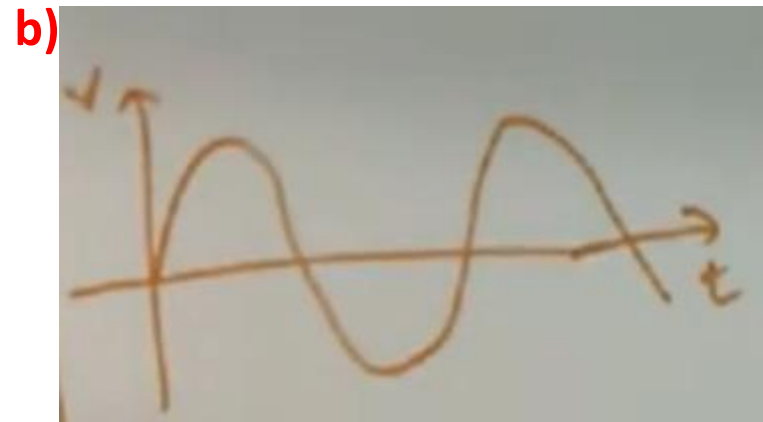
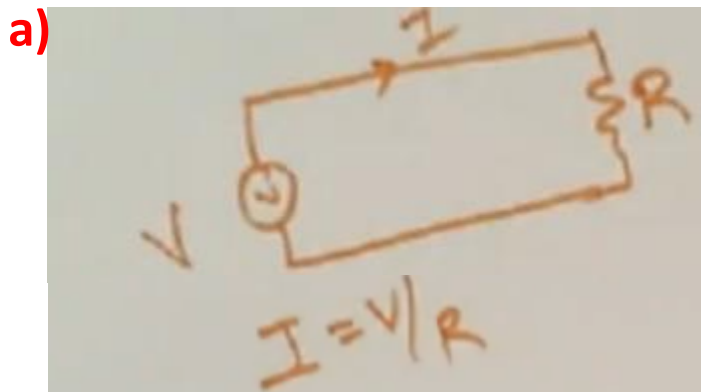
Typical transmission lines in cross-sectional view: (a) coaxial line, (b) two-wire line, (c) planar line, (d) wire above conducting plane, (e) microstrip line.

Example

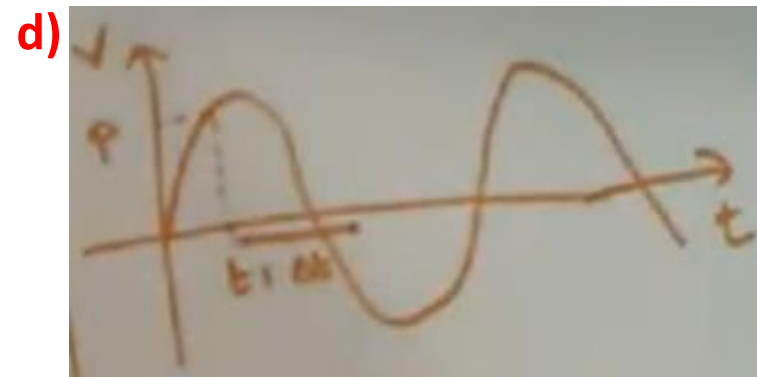


A transmission line can be defined as a device for propagating or guiding energy from one point to another.

Transit time



$$V - \alpha - IR - \alpha = 0$$



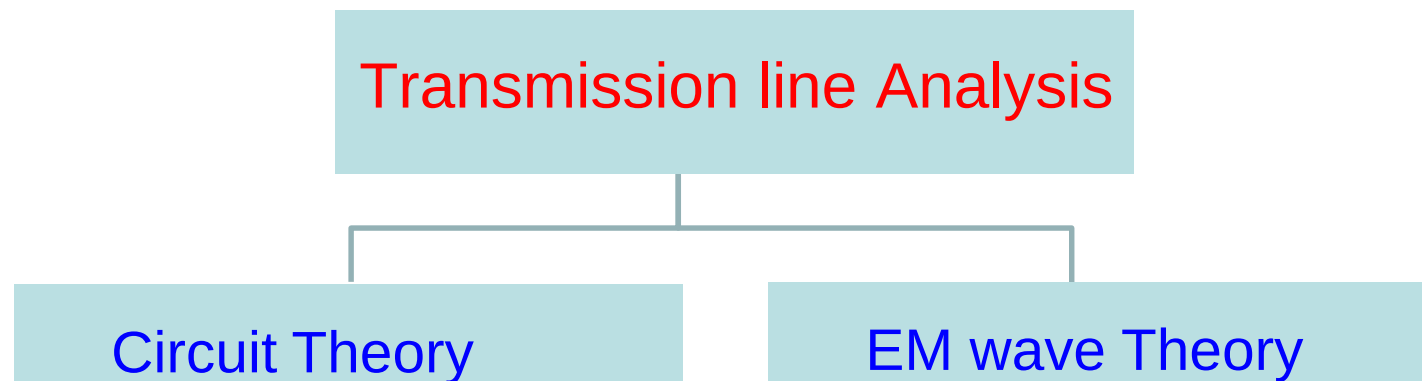
Value of Δt plays important role

High Frequency and longer lines : Circuit theory **X**

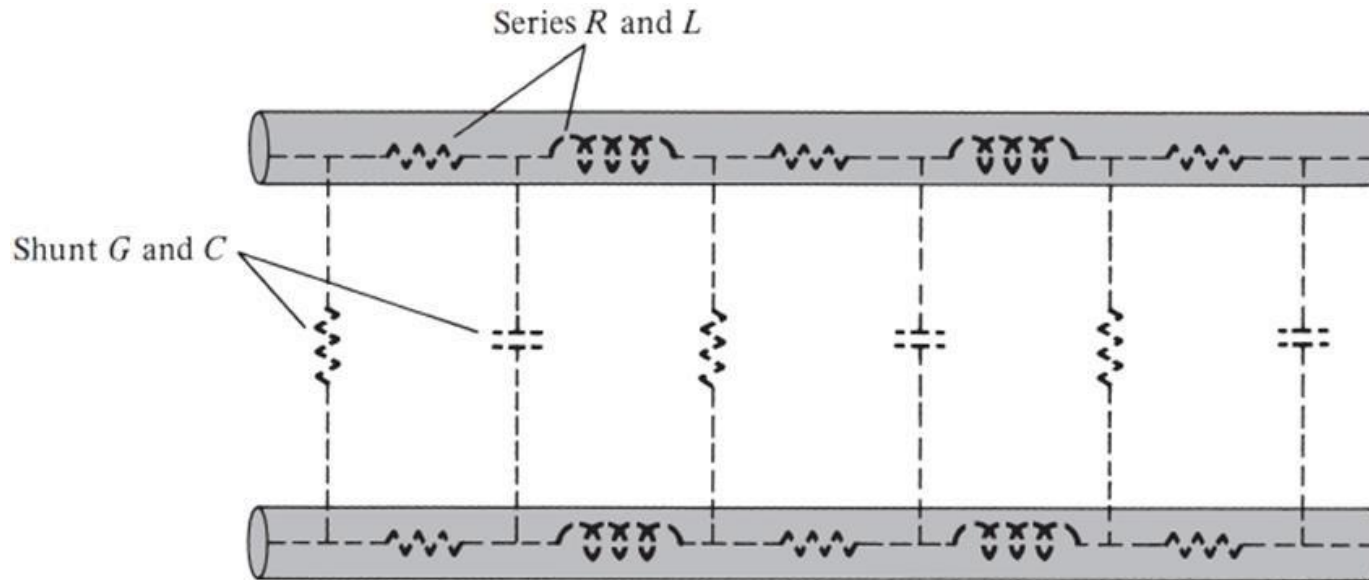
Response of parameters wrt frequency



- **At low frequencies**, the circuit elements are lumped since voltage and current waves affect the entire circuit at the same time.
- **At microwave frequencies**, such treatment of circuit elements is not possible since voltage and current waves do not affect the entire circuit at the same time.



Distributed parameters of a two-conductor transmission line



The impedance parameters L , R , C , and G represent:

L = series inductance per unit length

R = series resistance per unit length

C = shunt capacitance per unit length

G = shunt conductance per unit length.

General Transmission-line Equations



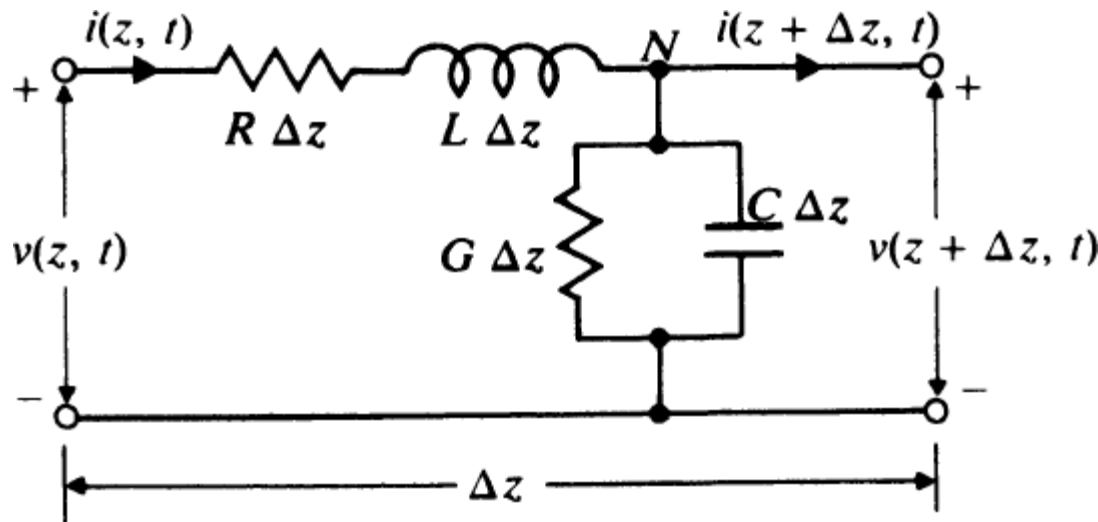
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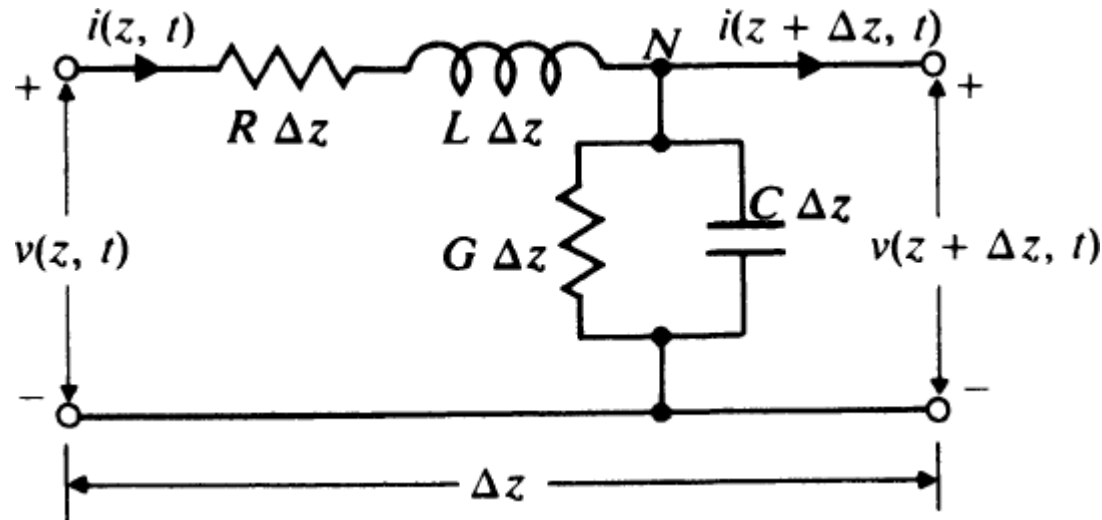
R = series resistance per unit length

C = shunt capacitance per unit length

G = shunt conductance per unit length.



General Transmission-line Equations

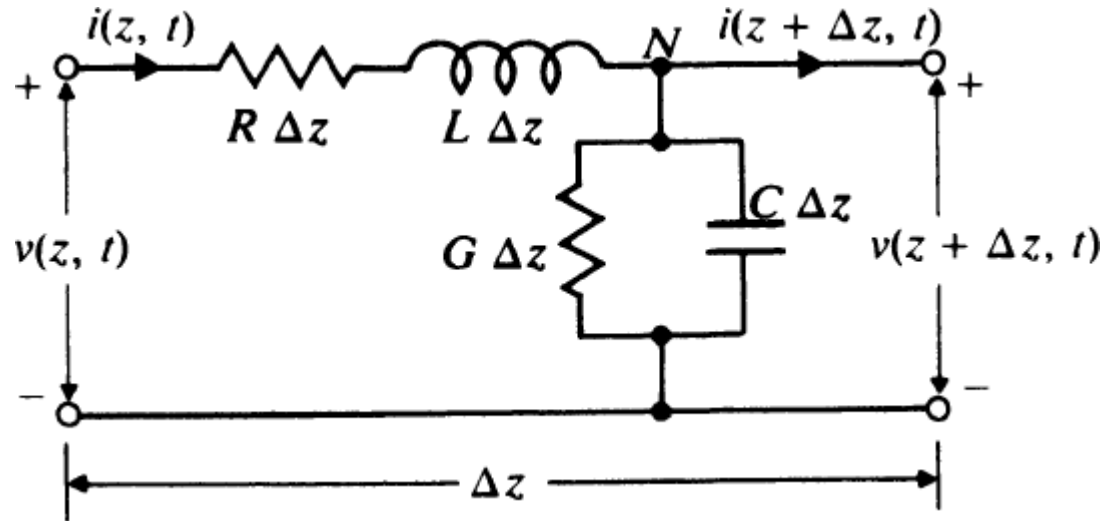


Kirchhoff's voltage law to the outer loop

$$V(z, t) = R \Delta z I(z, t) + L \Delta z \frac{\partial I(z, t)}{\partial t} + V(z + \Delta z, t)$$

$$\frac{V(z + \Delta z, t) - V(z, t)}{\Delta z} = R I(z, t) + L \frac{\partial I(z, t)}{\partial t}$$

General Transmission-line Equations



Kirchoff's current law

$$\begin{aligned}
 I(z, t) &= I(z + \Delta z, t) + \Delta I \\
 &= I(z + \Delta z, t) + G \Delta z V(z + \Delta z, t) + C \Delta z \frac{\partial V(z + \Delta z, t)}{\partial t}
 \end{aligned}$$

or

$$\frac{I(z + \Delta z, t) - I(z, t)}{\Delta z} = G V(z + \Delta z, t) + C \frac{\partial V(z + \Delta z, t)}{\partial t}$$

General Transmission-line Equations



$$-\frac{\partial v(z, t)}{\partial z} = Ri(z, t) + L \frac{\partial i(z, t)}{\partial t}$$

$$-\frac{\partial i(z, t)}{\partial z} = Gv(z, t) + C \frac{\partial v(z, t)}{\partial t}$$

General
Transmission-line
Equations

For Time-Harmonic Dependence:

$$V(z, t) = \text{Re} [V_s(z) e^{j\omega t}]$$

$$I(z, t) = \text{Re} [I_s(z) e^{j\omega t}]$$

$$-\frac{dV_s}{dz} = (R + j\omega L) I_s$$

$$-\frac{dI_s}{dz} = (G + j\omega C) V_s$$

Across the line,
values of V and I
are different

Time-Harmonic
Transmission-line
Equations

General Transmission-line Equations

$$\begin{aligned}-\frac{dV(z)}{dz} &= (R + j\omega L)I(z), \\ -\frac{dI(z)}{dz} &= (G + j\omega C)V(z).\end{aligned}$$

$$\gamma = \alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)} \quad (m^{-1})$$

$$\frac{d^2 V(z)}{dz^2} = \gamma^2 V(z)$$

$$\frac{d^2 I(z)}{dz^2} = \gamma^2 I(z),$$

Wave characteristic of an infinite Transmission line



The phasor voltage and current wave equations may be written as

$$\frac{d^2 V(z)}{dz^2} = \gamma^2 V(z)$$

$$\frac{d^2 I(z)}{dz^2} = \gamma^2 I(z),$$

The general solutions to the voltage and current wave equations are

$$\begin{aligned} V(z) &= V^+(z) + V^-(z) \\ &= V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z}, \end{aligned}$$

$$\begin{aligned} I(z) &= I^+(z) + I^-(z) \\ &= I_0^+ e^{-\gamma z} + I_0^- e^{\gamma z}, \end{aligned}$$

+z-directed waves

-z-directed waves

Propagation constant:

$$\gamma = \alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)} \quad (m^{-1})$$

Characteristic impedance



$$-\frac{dV(z)}{dz} = (R + j\omega L)I(z),$$

$$-\frac{dI(z)}{dz} = (G + j\omega C)V(z).$$

$$\begin{aligned} V(z) &= V^+(z) + V^-(z) \\ &= V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z}, \end{aligned}$$

$$\begin{aligned} I(z) &= I^+(z) + I^-(z) \\ &= I_0^+ e^{-\gamma z} + I_0^- e^{\gamma z}, \end{aligned}$$

$$\frac{d}{dz} (V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z}) = -(R + j\omega L) (I_0^+ e^{-\gamma z} + I_0^- e^{\gamma z})$$

$$\underbrace{V_0^+ (-\gamma) e^{-\gamma z}} + \underbrace{(\gamma) V_0^- e^{\gamma z}} = -\underbrace{(R + j\omega L) I_0^+ e^{-\gamma z}} - \underbrace{(R + j\omega L) I_0^- e^{\gamma z}}$$

$$\frac{V_0^+}{I_0^+} = \frac{(R + j\omega L)}{\gamma} = -\frac{V_0^-}{I_0^-} \Rightarrow Z_0$$

$$-\underbrace{(R + j\omega L) I_0^- e^{\gamma z}}$$

Wave characteristic of an infinite Transmission line



Characteristic impedance:

The **characteristic impedance** Z_o of the line is the ratio of positively traveling voltage wave to current wave at any point on the line.

$$Z_o = \frac{V_o^+}{I_o^+} = -\frac{V_o^-}{I_o^-} = \frac{R + j\omega L}{\gamma} = \frac{\gamma}{G + j\omega C}$$

$$= \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

Propagation constant:

$$\gamma = \alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)} \quad (m^{-1})$$

Secondary Constants:

Propagation constant

Characteristic impedance

Classification of Transmission lines



➤ **Lossless line** ($R = 0$ and $G = 0$)

A **transmission line** is said to be **lossless** if the conductors of the line are perfect ($\sigma_c \approx \infty$) and the dielectric medium separating them is lossless ($\sigma \approx 0$).

$$\alpha = 0, \quad \gamma = j\beta = j\omega \sqrt{LC}$$

$$u = \frac{\omega}{\beta} = \frac{1}{\sqrt{LC}} = f\lambda$$

$$X_o = 0, \quad Z_o = R_o = \sqrt{\frac{L}{C}}$$

Wave characteristic of Transmission line (lossless)



$$V(z) = V^+(z) + V^-(z) \\ = \boxed{V_0^+ e^{-\gamma z}} + V_0^- e^{\gamma z},$$

[For Time harmonic variation]

$$v(z) = V_0^+ e^{-\gamma z} e^{j\omega t} \\ = V_0^+ e^{-\alpha z} e^{-j\beta z} e^{j\omega t}$$

For lossless line $\alpha = 0$

$$V(z) = V_0^+ e^{j(\omega t - \beta z)} \text{ or } V_0^+ e^{-j(\beta z - \omega t)}$$

$$\underline{\underline{\text{Re}[v(z)] = V_0^+ \cos(\beta z - \omega t)}} \quad \text{Wave nature}$$

Wave characteristic of Transmission line (not lossless)



$$V(z) = V^+(z) + V^-(z) \\ = \underline{V_0^+ e^{-\gamma z}} + V_0^- e^{\gamma z},$$

$$v(z) = V_0^+ e^{-\alpha^2 - j\beta^2} e^{j\omega t}$$

Time harmonic e

$$v(z) = V_0^+ e^{-\alpha z} e^{-j\beta z - \omega t}$$

What is $\beta \rightarrow ??$

units

What is $\alpha \rightarrow ??$

units

$$e^{-\alpha^2} \Rightarrow$$

$$[1 \text{ Np/m} = -8.68 \text{ dB}] ??$$

Classification of Transmission lines



➤ Distortionless line ($R/L=G/C$)

A **distortionless line** is one in which the attenuation constant α is frequency independent while the phase constant β is linearly dependent on frequency.

$$\begin{aligned}\gamma &= \sqrt{RG \left(1 + \frac{j\omega L}{R}\right) \left(1 + \frac{j\omega C}{G}\right)} \\ &= \sqrt{RG} \left(1 + \frac{j\omega C}{G}\right) = \alpha + j\beta\end{aligned}$$

$$\alpha = \sqrt{RG}, \quad \beta = \omega \sqrt{LC}$$

$$Z_o = \sqrt{\frac{R(1 + j\omega L/R)}{G(1 + j\omega C/G)}} = \sqrt{\frac{R}{G}} = \sqrt{\frac{L}{C}} = R_o + jX_o$$

$$R_o = \sqrt{\frac{R}{G}} = \sqrt{\frac{L}{C}}, \quad X_o = 0$$

$$u = \frac{\omega}{\beta} = \frac{1}{\sqrt{LC}} = f\lambda$$

Classification of Transmission lines



Low loss line ($R \ll \omega L$, $G \ll \omega C$)

$$\begin{aligned}\gamma &= \alpha + j\beta = j\omega\sqrt{LC}\left(1 + \frac{R}{j\omega L}\right)^{1/2}\left(1 + \frac{G}{j\omega C}\right)^{1/2} \\ &\cong j\omega\sqrt{LC}\left(1 + \frac{R}{2j\omega L}\right)\left(1 + \frac{G}{2j\omega C}\right) \\ &\cong j\omega\sqrt{LC}\left[1 + \frac{1}{2j\omega}\left(\frac{R}{L} + \frac{G}{C}\right)\right];\end{aligned}$$

$$\alpha \cong \frac{1}{2}\left(R\sqrt{\frac{C}{L}} + G\sqrt{\frac{L}{C}}\right)$$

$$\beta \cong \omega\sqrt{LC} \quad (\text{approximately a linear function of } \omega).$$

Classification of Transmission lines



Low loss line ($R \ll \omega L$, $G \ll \omega C$)

$$\alpha_p = \frac{\omega}{\beta} \cong \frac{1}{\sqrt{LC}}$$

$$Z_0 = R_0 + jX_0 = \sqrt{\frac{L}{C}} \left(1 + \frac{R}{j\omega L}\right)^{1/2} \left(1 + \frac{G}{j\omega C}\right)^{-1/2}$$

$$\cong \sqrt{\frac{L}{C}} \left[1 + \frac{1}{2j\omega} \left(\frac{R}{L} - \frac{G}{C}\right)\right];$$

$$R_0 \cong \sqrt{\frac{L}{C}}$$

$$X_0 \cong -\sqrt{\frac{L}{C}} \frac{1}{2\omega} \left(\frac{R}{L} - \frac{G}{C}\right)$$

Reflection coefficient



The general solutions to the voltage and current wave equations are

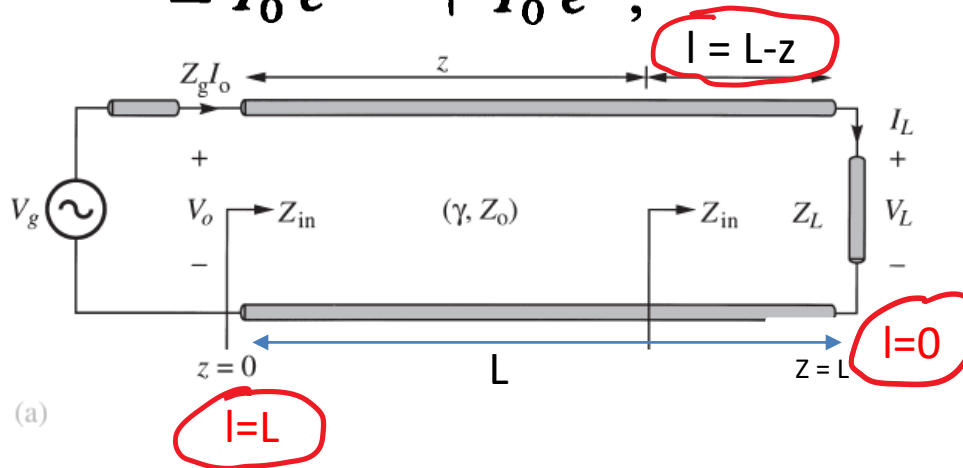
$$V(z) = V^+(z) + V^-(z)$$

$$I(z) = I^+(z) + I^-(z) \quad ;$$

$$= I_0^+ e^{-\gamma z} + I_0^- e^{\gamma z},$$

$$V_s(z) = V_o^+ e^{-\gamma z} + V_o^- e^{\gamma z}$$

$$I_s(z) = \frac{V_o^+}{Z_o} e^{-\gamma z} - \frac{V_o^-}{Z_o} e^{\gamma z}$$



$$Z_{in} = \frac{V_s(z)}{I_s(z)} = Z_o \frac{V_o^+ e^{-\gamma z} + V_o^- e^{\gamma z}}{V_o^+ e^{-\gamma z} - V_o^- e^{\gamma z}}$$

**Voltage reflection coefficient
(at load, l)**

$$\Gamma(l) = \frac{V_o^- e^{-\gamma l}}{V_o^+ e^{\gamma l}} \approx \frac{V_o^-}{V_o^+} e^{-2\gamma l} = \text{Reflection coefficient}$$

$$\left[\begin{aligned} V_s(l) &= V_o^+ e^{-\gamma l} + V_o^- e^{\gamma l} \\ I_s(l) &= \frac{V_o^+}{Z_o} e^{-\gamma l} - \frac{V_o^-}{Z_o} e^{\gamma l} \end{aligned} \right]$$

Reflection coefficient



$$\Gamma(l) = \frac{V_o^- e^{-\gamma l}}{V_o^+ e^{\gamma l}} \approx \frac{V_o^-}{V_o^+} e^{-2\gamma l} = \text{Reflection coefficient}$$

at $l=0$ $\left[\Gamma_{l=0} = \frac{V_o^-}{V_o^+} \right]$

So $\left[\Gamma(l) = (\Gamma_{l=0}) e^{-2\gamma l} \right] \Rightarrow \underline{\text{Important}}$

$$\left[\begin{aligned} V_s(l) &= V_o^+ e^{+\gamma l} + V_o^- e^{-\gamma l} \\ I_s(l) &= \frac{V_o^+}{Z_o} e^{+\gamma l} - \frac{V_o^-}{Z_o} e^{-\gamma l} \end{aligned} \right] \text{ So } Z(l) = \frac{V_s(l)}{I_s(l)}$$

$$\left[Z(l) = \left(\frac{V_o^+ e^{+\gamma l} + V_o^- e^{-\gamma l}}{V_o^+ e^{+\gamma l} - V_o^- e^{-\gamma l}} \right) Z_o \right]$$

Impedance on any point of transmission line



$$Z(l) = \frac{V_S(l)}{I_S(l)}$$

$$\left[Z(l) = \frac{V_0^+ e^{\gamma l} + V_0^- e^{-\gamma l}}{V_0^+ e^{\gamma l} - V_0^- e^{-\gamma l}} Z_0 \right] \Rightarrow Z(l) = Z_0 \frac{V_0^+ e^{\gamma l} \left[1 + \frac{V_0^-}{V_0^+} e^{-2\gamma l} \right]}{V_0^+ e^{\gamma l} \left[1 - \frac{V_0^-}{V_0^+} e^{-2\gamma l} \right]}$$

$$S. \quad Z(l) = Z_0 \frac{[1 + \Gamma(l)]}{[1 - \Gamma(l)]}$$

$$Z(l) = Z_0 \frac{[1 + \Gamma(l)]}{[1 - \Gamma(l)]}$$

$$\Gamma(l) = \frac{Z(l) - Z_0}{Z(l) + Z_0}$$

Imp. at load $l=0$

$$\Gamma(0) = \frac{Z_L - Z_0}{Z_L + Z_0}$$

Characteristics of transmission line



$$Z(l) = Z_0 \left[\frac{1 + \Gamma(l)}{1 - \Gamma(l)} \right] \quad \cdot \quad \Gamma(l) = \frac{V_0^-}{V_0^+} e^{-2\gamma l}$$

$$\Gamma(l) = \frac{Z(l) - Z_0}{Z(l) + Z_0}$$

$$\Gamma_{l(l=0)} = \frac{V_0^-}{V_0^+}$$

$$\Gamma_{l \text{ at } l=0} = \frac{Z_L - Z_0}{Z_L + Z_0}$$

at $l=0$

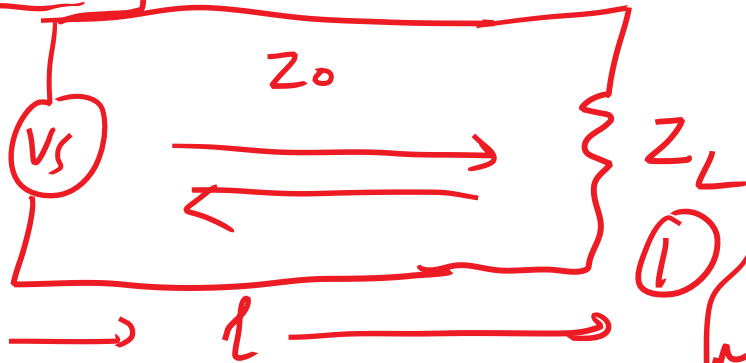
$$Z_L = \frac{Z_0(1 + \Gamma_l)}{(1 - \Gamma_l)}$$

$$0 \leq |\Gamma| \leq 1$$

Characteristics of transmission line



Cases:



$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} \quad Z_L = Z_0$$

$$\Gamma_L = 0 \quad \checkmark$$

No reflection

matched line

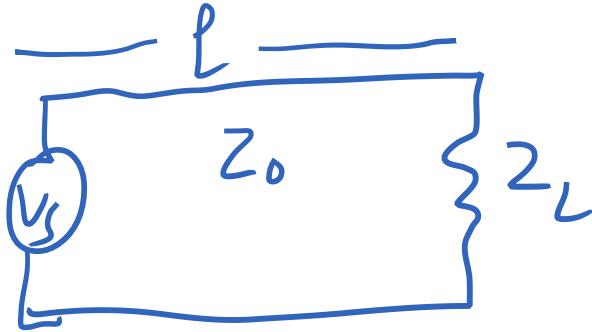
② Infinite line

$$\Gamma(l) = \Gamma_L e^{-2\gamma l}$$

$$= \Gamma_L e^{-2\alpha l - j\beta l}$$

There will be no reflection

Wave characteristic of finite Transmission line



$$Z(l) = Z_0 \left[\frac{1 + \Gamma(l)}{1 - \Gamma(l)} \right]$$

$$\Gamma(l) = \Gamma_0 e^{-2\gamma l}$$

$$\Gamma_0 = \left(\frac{Z_L - Z_0}{Z_L + Z_0} \right)$$

$$Z(l) = Z_0 \left[\frac{1 + \frac{Z_L - Z_0}{Z_L + Z_0} e^{-2\gamma l}}{1 - \frac{Z_L - Z_0}{Z_L + Z_0} e^{-2\gamma l}} \right]$$

Wave characteristic of finite Transmission line



$$Z(l) = Z_0 \left[\frac{1 + \frac{Z_L - Z_0}{Z_L + Z_0} e^{-2\gamma l}}{1 - \frac{Z_L - Z_0}{Z_L + Z_0} e^{-2\gamma l}} \right]$$

$$Z(l) = Z_0 \left[\frac{(Z_L + Z_0) + (Z_L - Z_0) e^{-2\gamma l}}{(Z_L + Z_0) - (Z_L - Z_0) e^{-2\gamma l}} \right]$$

$$Z(l) = Z_0 \left[\frac{Z_L (e^{\gamma l} + e^{-\gamma l}) + Z_0 (e^{\gamma l} - e^{-\gamma l})}{Z_L (e^{\gamma l} - e^{-\gamma l}) + Z_0 (e^{\gamma l} + e^{-\gamma l})} \right]$$

Wave characteristic of finite Transmission line



$$Z(l) = Z_0 \left[\frac{Z_L (e^{\gamma L} + e^{-\gamma L}) + Z_0 (e^{\gamma L} - e^{-\gamma L})}{Z_L (e^{\gamma L} - e^{-\gamma L}) + Z_0 (e^{\gamma L} + e^{-\gamma L})} \right]$$

$$Z(l) = Z_0 \left[\frac{Z_L \cosh(\gamma L) + Z_0 \sinh(\gamma L)}{Z_0 \cosh(\gamma L) + Z_L \sinh(\gamma L)} \right]$$

$$Z(l) = Z_0 \left[\frac{Z_L + Z_0 \tanh(\gamma L)}{Z_0 + Z_L \tanh(\gamma L)} \right] \text{ Imp.}$$

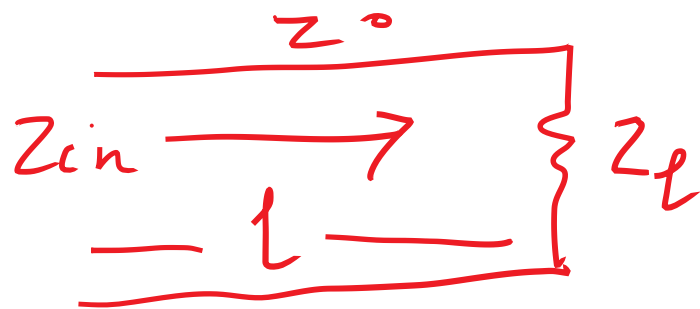
Input impedance at any point along a general (lossy) transmission line

Wave characteristic of finite Transmission line



$$Z_{in} = Z_o \left[\frac{Z_L + jZ_o \tan \beta l}{Z_o + jZ_L \tan \beta l} \right]$$

for lossless transmission line $\alpha = 0$



$$\text{So } \Gamma_L = \frac{Z_L - Z_o}{Z_L + Z_o}$$

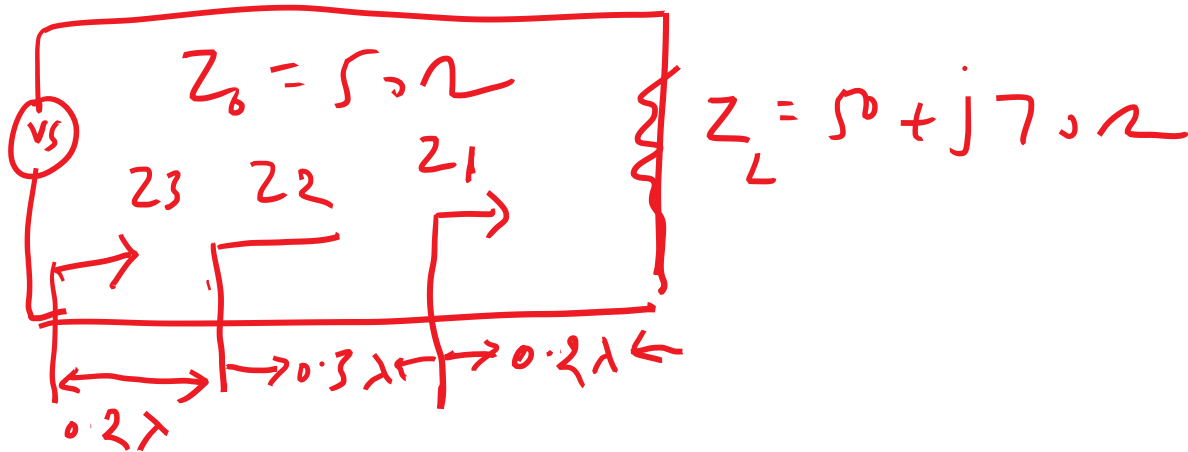
βl ??

$$Z(l) = Z_o \left[\frac{1 + \Gamma(l)}{1 - \Gamma(l)} \right]$$

$$\Gamma(l) = \Gamma_L e^{-2\gamma l}$$

$$Z_{in} = Z_o \left[\frac{Z_L + Z_o \tanh \gamma l}{Z_o + Z_L \tanh \gamma l} \right]$$

Example-1



Find out $Z_1 = ?$

$Z_2 = ?$

$Z_3 = ?$

Reflection Coefficient & VSWR



$$\Gamma(l) = \frac{Z(l) - Z_0}{Z(l) + Z_0} \quad \text{at } l=0$$
$$\Gamma_l = \frac{Z_l - Z_0}{Z_l + Z_0}$$
$$\Gamma_l = \frac{V_0^-}{V_0^+} \quad \Gamma(l) = \frac{V_0^-}{V_0^+} e^{-2\gamma l}$$

Measure of the mismatch of a line can be done by calculating standing wave ratio (SWR)-s

$$S = \frac{|V_{\max}|}{|V_{\min}|} = \frac{1 + |\Gamma|}{1 - |\Gamma|} \quad (\text{Dimensionless}).$$

$$|\Gamma| = \frac{S - 1}{S + 1} \quad (\text{Dimensionless}).$$

Reflection Coefficient & VSWR



$$\Gamma = \frac{Z_L - Z_o}{Z_L + Z_o}$$

$$|\Gamma| = \frac{S - 1}{S + 1}$$

$$\left[S = \frac{1 + |\Gamma|}{1 - |\Gamma|} \right]$$

$\Gamma = 0, \quad S = 1$ when $Z_L = Z_o$ (Matched load);

$\Gamma = -1, \quad S \rightarrow \infty$ when $Z_L = 0$ (Short circuit);

$\Gamma = +1, \quad S \rightarrow \infty$ when $Z_L \rightarrow \infty$ (Open circuit).

what is value of $|\Gamma|$??
if load is inductive or capacitive

$$V = V^+ e^{j\beta l} + V^- e^{-j\beta l} \text{ lossless line}$$

$$V = V^+ e^{j\beta l} + V^- e^{-j\beta l}$$

$$V = V^+ e^{j\beta l} \left(1 + \frac{V^-}{V^+} e^{-j2\beta l} \right)$$

$$V = V^+ e^{j\beta l} \left(1 + \Gamma_L e^{-j2\beta l} \right)$$

$$V = V^+ e^{j\beta l} \left(1 + |\Gamma_L| e^{j\phi} e^{-j2\beta l} \right)$$

$$|V| = |V^+| \left| 1 + |\Gamma_L| e^{j(\phi - 2\beta l)} \right|$$

$$\text{for } V_{\max} \rightarrow |e^{j(\phi - 2\beta l)}| = 1$$

$I_{\min}$

$$\text{For } V_{\min}: \\ |e^{j(\phi - 2\beta l)}| = -1$$

$$\phi - 2\beta l = (2n+1)\pi$$

$I_{\max}$

$$\text{So } \phi - 2\beta l = 2n\pi$$

Rmax and Rmin



$$V_{max} = V^+ (1 + \Gamma_L)$$

$$I_{min} = \frac{V^+}{Z_0} (1 - \Gamma_L)$$

$$V_{min} = V^+ (1 - \Gamma_L)$$

$$I_{max} = \frac{V^+}{Z_0} (1 + \Gamma_L)$$

$$\therefore R_{max} = Z_0 \left(\frac{1 + \Gamma_L}{1 - \Gamma_L} \right)$$

$$R_{min} = \frac{Z_0}{\left(\frac{1 + \Gamma_L}{1 - \Gamma_L} \right)}$$

$$\therefore \underline{R_{max} = Z_0 (VSWR)}$$

$$R_{min} = \frac{Z_0}{VSWR}$$

Example



$$Z_{in} = \frac{Z_0 (Z_L + j Z_0 \tan \beta L)}{(Z_0 + j Z_L \tan \beta L)}$$

a) find Z_{in} at $l = \frac{\lambda}{2}$ and $l = \frac{\lambda}{4}$?

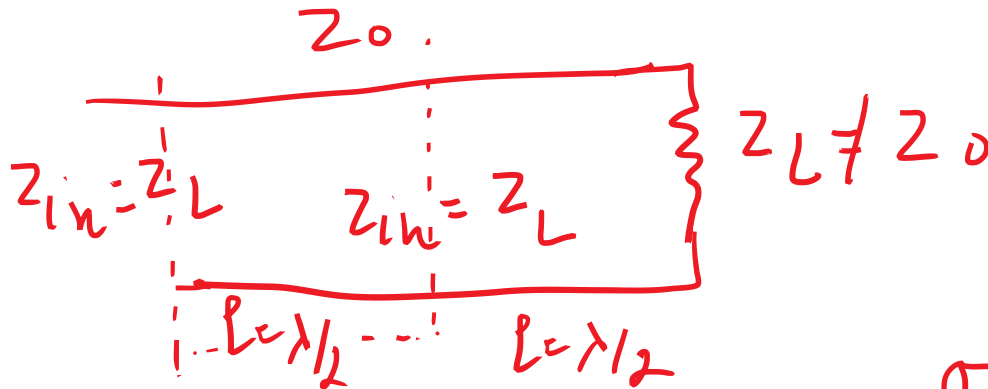
at $l = \lambda/2$

$$Z_{in} = Z_L$$

at $l = \lambda/4$

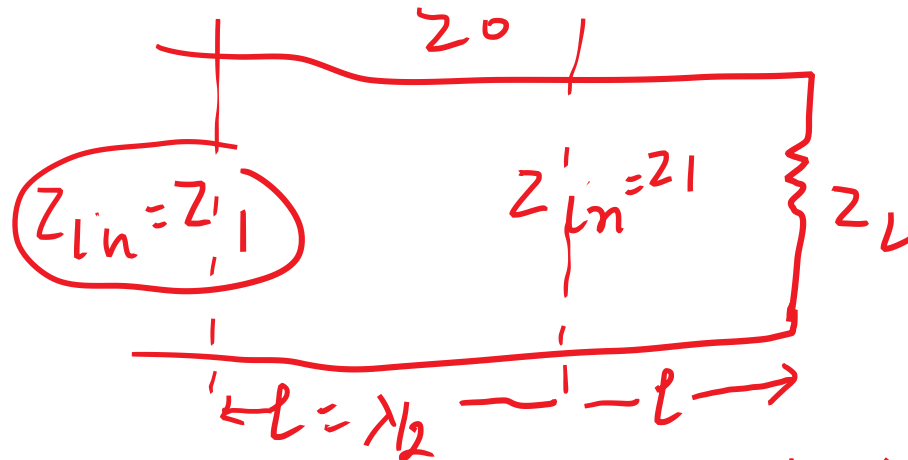
$$Z_{in} = \frac{Z_0^2}{Z_L}$$

Example



at $l = \lambda/2$

$$Z_{in} = Z_L$$



$\Rightarrow$ Impedance repeats at every distance $= \frac{\lambda}{2}$

Example-2



$$Z_0 = 50 \Omega$$

$$Z_L = 75 - 75j \Omega$$

Calculate a) Γ_L , VSWR

$$\text{VSWR} = 3.34$$

$$\Gamma_L = 0.54 \angle -40.6^\circ$$

$$\Gamma(0.2\lambda) = 0.54 \angle +175.4^\circ$$

$$Z(0.2\lambda) = (14.95 + j1.82) \Omega$$

$$\Gamma(0.3\lambda) = 0.54 \angle 103.39^\circ$$

$$Z(0.3\lambda) = (22.83 + j34.136) \Omega$$

b) $Z_{in}(0.2\lambda)$, $\Gamma_{in}(0.2\lambda)$

c) $Z_{in}(0.3\lambda)$, $\Gamma_{in}(0.3\lambda)$

d) $Z_{in}(0.7\lambda)$, $\Gamma_{in}(0.7\lambda)$

e) Location of V_{max} & V_{min} -

Example-2



Location of $V_{\max} \approx I_{\min} \approx R_{\max}$

$$\phi - 2\beta l = 2n\pi \quad n=1$$

$$-40.6^\circ - 2\beta l = 360^\circ$$

$$-2\beta l = 400.6^\circ$$

$$-l = \frac{400.6}{2 \times 2 \times 180} \lambda$$

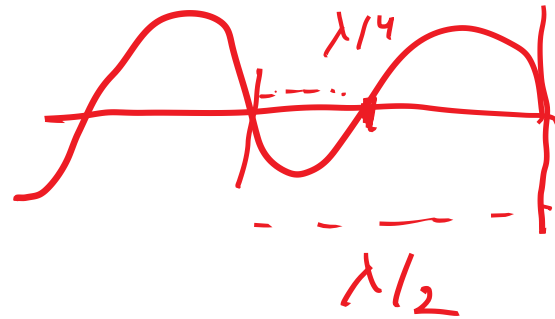
$$l = -0.55\lambda + 0.5\lambda + 0.5\lambda = 0.45\lambda$$

maxima ↑

minima

impedance repeats at every $\lambda/2$ length

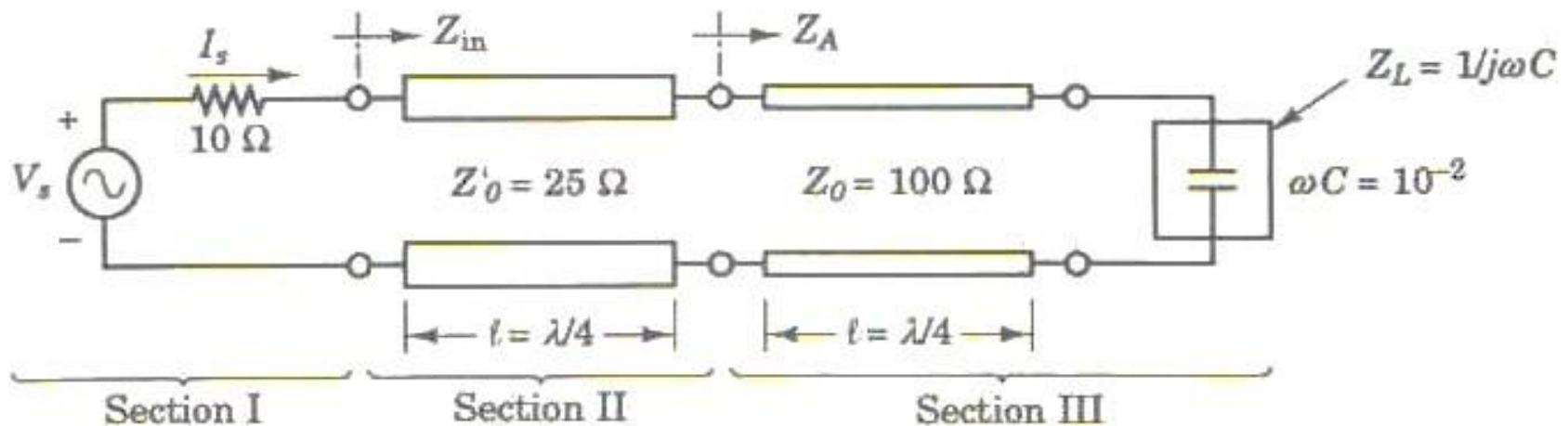
$$\rightarrow 0.45\lambda - 0.25\lambda = \boxed{0.20\lambda}$$

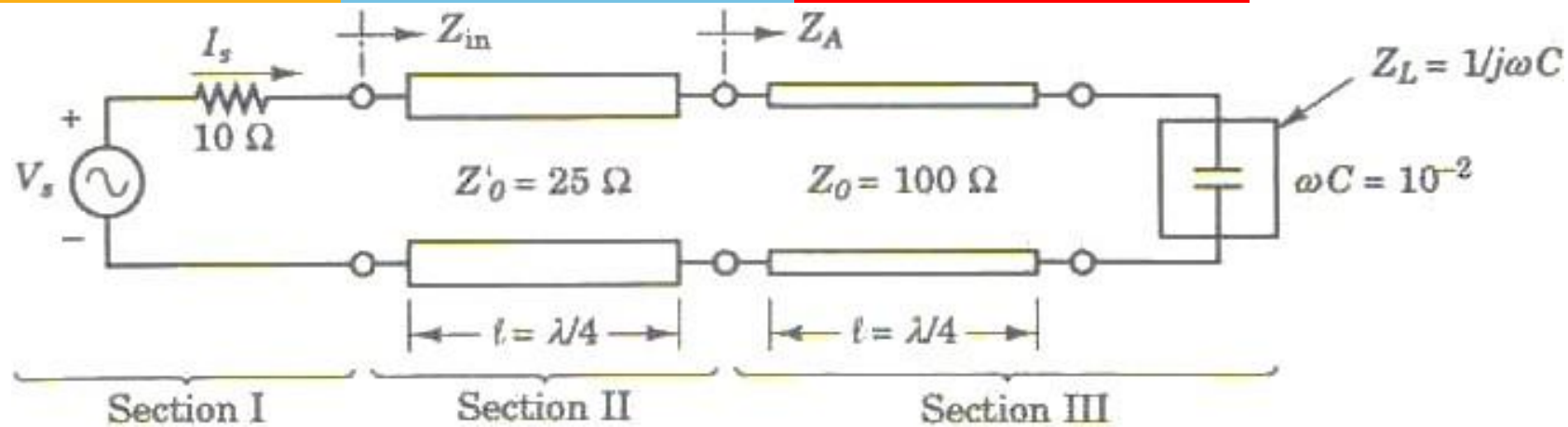


Example-2



What is the input impedance supplied by the source?





$$Z_i = Z_o \left[\frac{Z_L + jZ_o \tan \beta l}{Z_o + jZ_L \tan \beta l} \right]$$

$$Z_{\lambda/4} = \frac{Z_o^2}{Z_L}$$

Transmission Lines as circuit elements



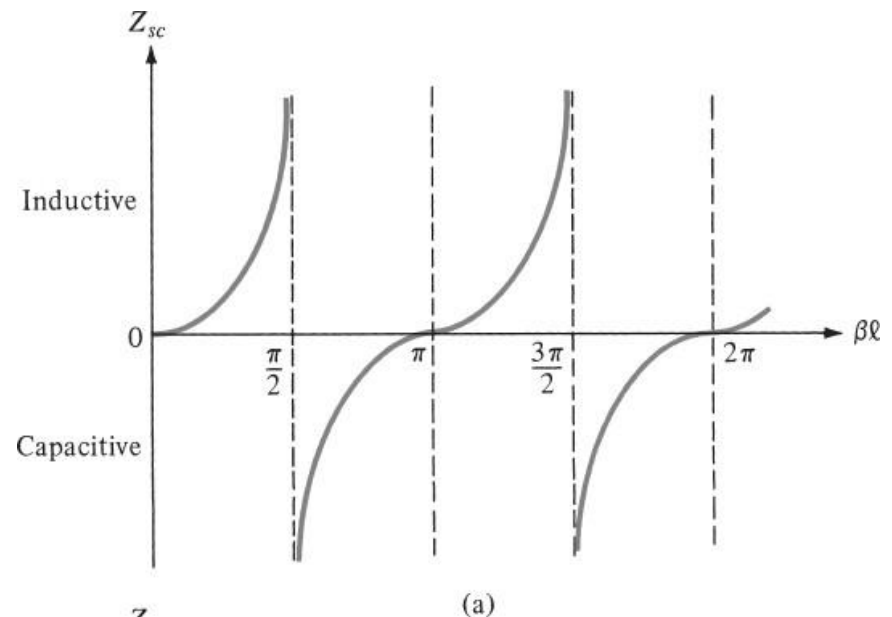
For a lossless line:

$$Z_{in} = Z_o \left[\frac{Z_L + jZ_o \tan \beta l}{Z_o + jZ_L \tan \beta l} \right]$$

Transmission Lines terminated in short or open circuits are commonly used as tuning elements.

If the line is **short-circuited**: $Z_L = 0$

$$Z_{SC} = jZ_o \tan \beta l$$



Transmission Lines as circuit elements



$$Z_{sc} = jZ_0 \tan \beta l$$

$$\text{if } \beta l < \pi/2 \\ l < \frac{\lambda}{4}$$

$$\tan \beta l \approx \beta l$$

$$Z_{sc} = jZ_0 \beta l$$

$$Z_{sc} = j\sqrt{\frac{L}{C}} \omega \sqrt{LC} \cdot l = j\omega L l = \underbrace{j\omega L}_{\text{Inductor}} \Rightarrow$$

$$\pi/2 < \beta l < \pi \rightarrow \text{Capacitor}$$

$$\text{at } \beta l = \pi/2 \rightarrow Z_{sc} = \infty \rightarrow \text{capacitor} \checkmark$$

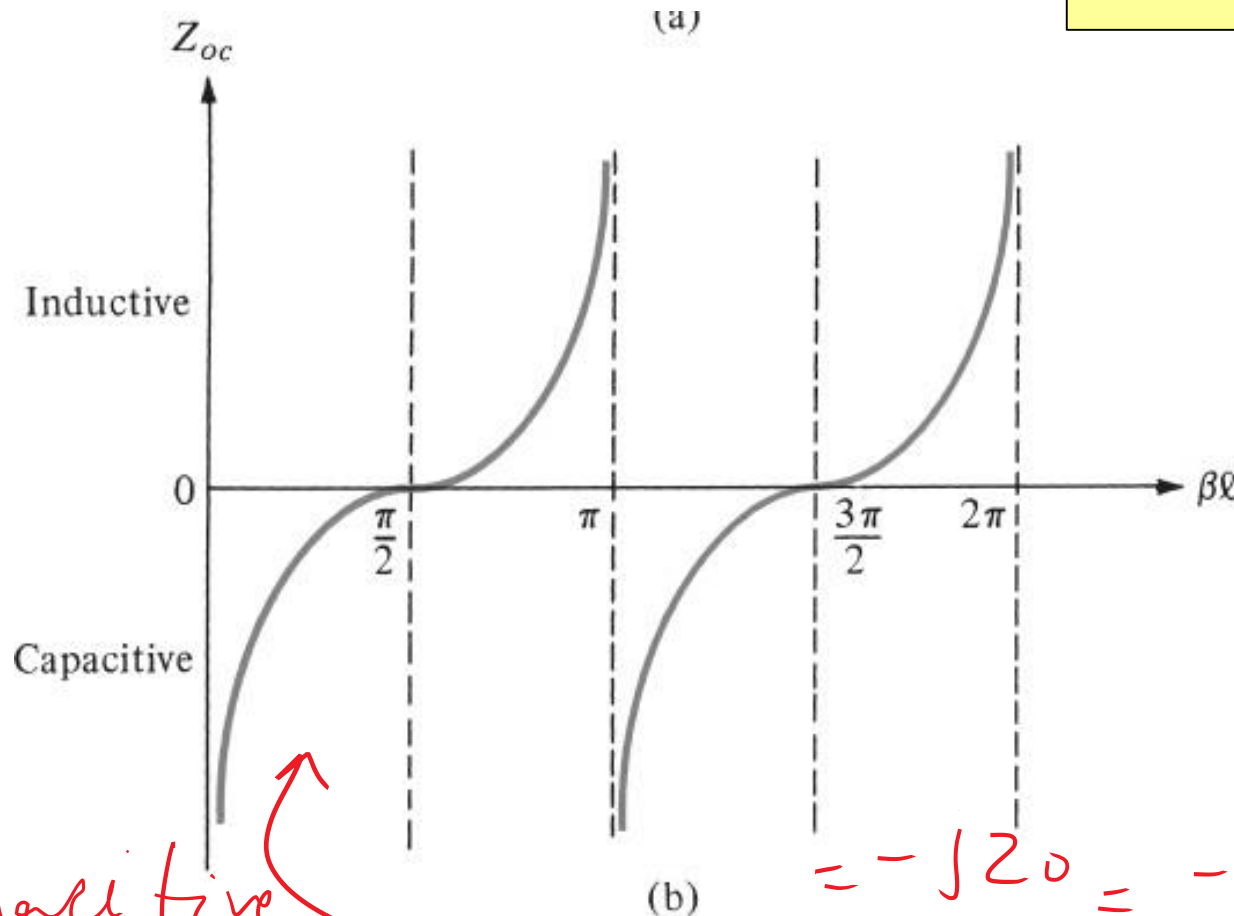
Transmission Lines as circuit elements



For a lossless line: open-circuited:

$$Z_i = Z_o \left\{ \frac{Z_L + jZ_o \tan \beta l}{Z_o + jZ_L \tan \beta l} \right\}$$

$$Z_{in} = -jZ_o \cot \beta l$$



$$\beta l < \frac{\pi}{2}$$

$$l < \frac{\lambda}{4}$$

$$= -jZ_o \tan \beta l$$

$$= -jZ_o = -j\sqrt{L/C} = \left[\frac{1}{j\omega C} \right]$$

capacitive

Transmission Lines as circuit elements



$$Z_{sc} = jZ_0 \tan \beta l$$

$$Z_{oc} = -jZ_0 \cot \beta l = \frac{Z_0}{j \tan \beta l}$$

$$Z_0 = \sqrt{Z_{oc} Z_{sc}}$$

$$Z_0 = \sqrt{Z_{OC} Z_{SC}}$$
$$\gamma = \frac{1}{l} \tanh^{-1} \sqrt{\frac{Z_{SC}}{Z_{OC}}}$$

$$\frac{Z_{sc}}{Z_{oc}} = \tan^2 \beta l$$

$$\beta = \frac{1}{l} \tan^{-1} \sqrt{\frac{Z_{sc}}{Z_{oc}}}$$

Lossy line

Smith Chart



Transmission line calculation-such as the determination of input impedance, reflection coefficient and load impedance often involve tedious manipulations of complex numbers.

Graphical method of solution → **Smith Chart**

Smith Chart was devised by P.H.Smith (in 1930).

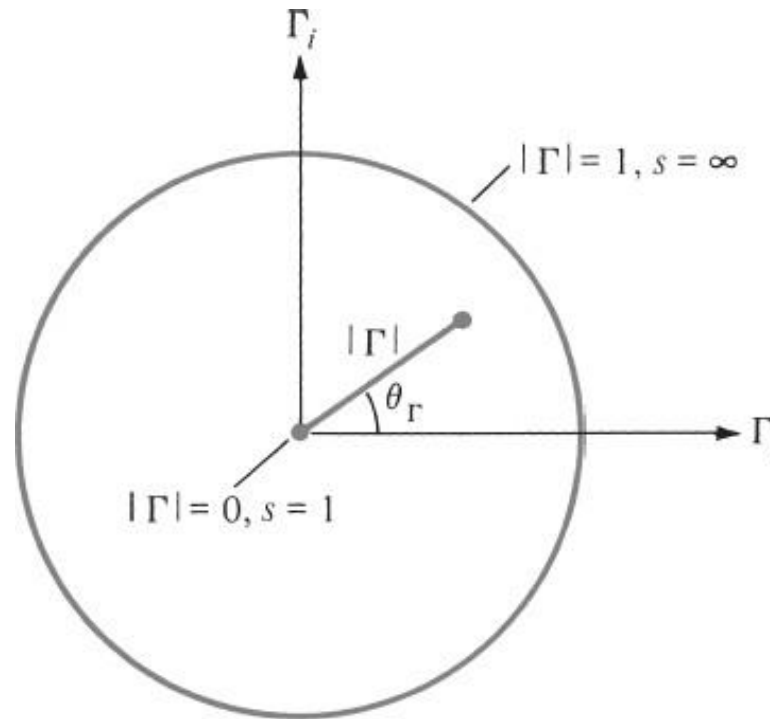
Smith Chart is a graphical plot of normalized resistance and reactance functions in the reflection-coefficient plane

Reflection Co-efficient plane (Finite plane)



$$\left[\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} \right]$$

Z_0 normalized impedance



What
about
Z impedance
plane??
==

The **normalized impedance** is represented on the **Smith chart** by using families of curves that identify the **normalized resistance r** (real part) and the **normalized reactance x** (imaginary part)

Smith Chart Circle Equations



- If a lossless line of Z_0 is terminated with Z_L ,
 $z_L = Z_L/Z_0$ (normalized load impedance),

$$\Gamma = \frac{z_L - 1}{z_L + 1} = |\Gamma| e^{j\theta} \quad \longrightarrow \quad z_L = \frac{1 + |\Gamma| e^{j\theta}}{1 - |\Gamma| e^{j\theta}}$$

- Let $\Gamma = \Gamma_r + j\Gamma_i$, and $z_L = r_L + jx_L$.

$$r_L + jx_L = \frac{(1 + \Gamma_r) + j\Gamma_i}{(1 - \Gamma_r) - j\Gamma_i} \quad \longrightarrow \quad r_L = \frac{1 - \Gamma_r^2 - \Gamma_i^2}{(1 - \Gamma_r)^2 + \Gamma_i^2} \quad x_L = \frac{2\Gamma_i}{(1 - \Gamma_r)^2 + \Gamma_i^2}$$

$$\left(\Gamma_r - \frac{r_L}{1 + r_L} \right)^2 + \Gamma_i^2 = \left(\frac{1}{1 + r_L} \right)^2$$
$$(\Gamma_r - 1)^2 + \left(\Gamma_i - \frac{1}{x_L} \right)^2 = \left(\frac{1}{x_L} \right)^2$$

r circle

x circle

Smith Chart Circle Equations



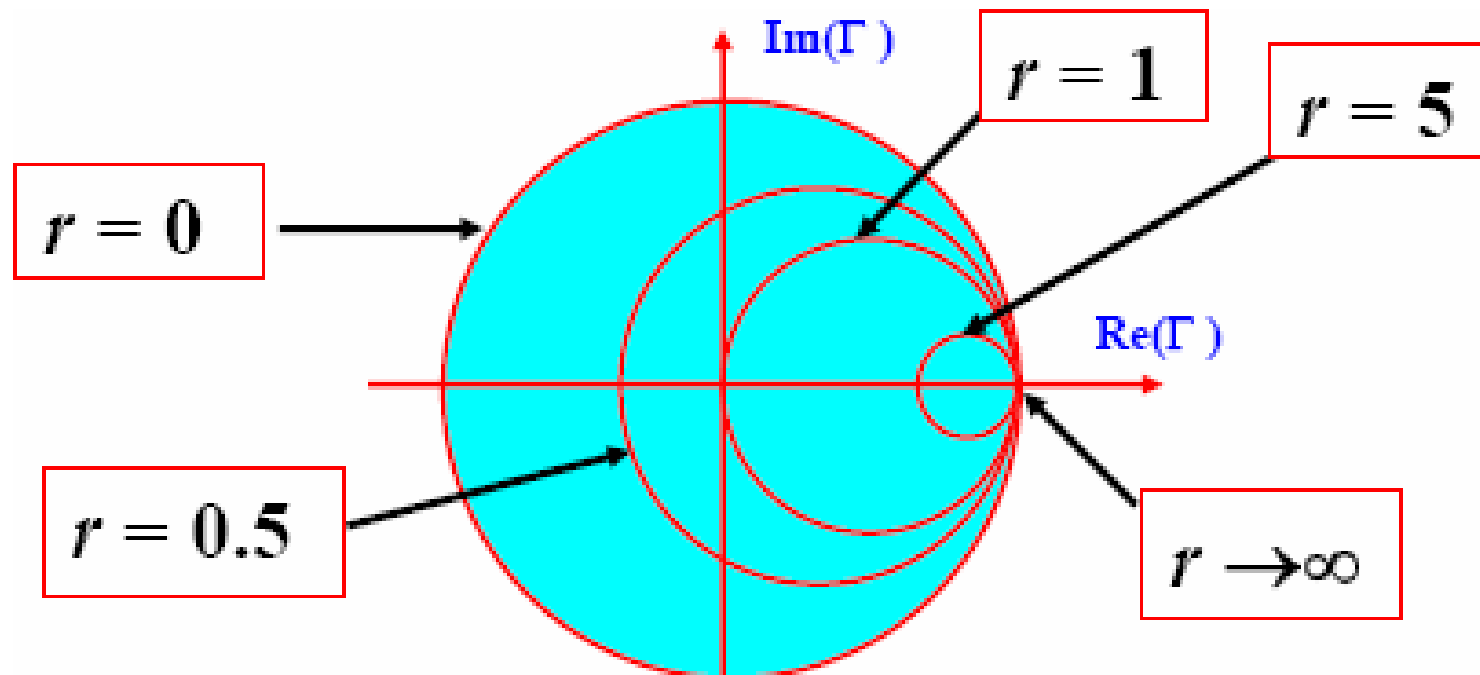
$$\left(\Gamma_r - \frac{r_L}{1 + r_L} \right)^2 + \Gamma_i^2 = \left(\frac{1}{1 + r_L} \right)^2$$
$$(\Gamma_r - 1)^2 + \left(\Gamma_i - \frac{1}{x_L} \right)^2 = \left(\frac{1}{x_L} \right)^2$$

The result for the **real part** indicates that on the complex plane with coordinates $(\text{Re}(\Gamma), \text{Im}(\Gamma))$ all the possible impedances with a given normalized resistance r are found on a **circle** with

$$\text{Center} = \left\{ \frac{r}{1 + r}, 0 \right\} \qquad \text{Radius} = \frac{1}{1 + r}$$

As the normalized resistance r varies from **0** to ∞ , we obtain a family of circles completely contained inside the domain of the reflection coefficient $|\Gamma| \leq 1$.

r-circles



$$\text{Center} = \left\{ \frac{r}{1+r}, 0 \right\}$$

$$\text{Radius} = \frac{1}{1+r}$$

Smith Chart Circle Equations



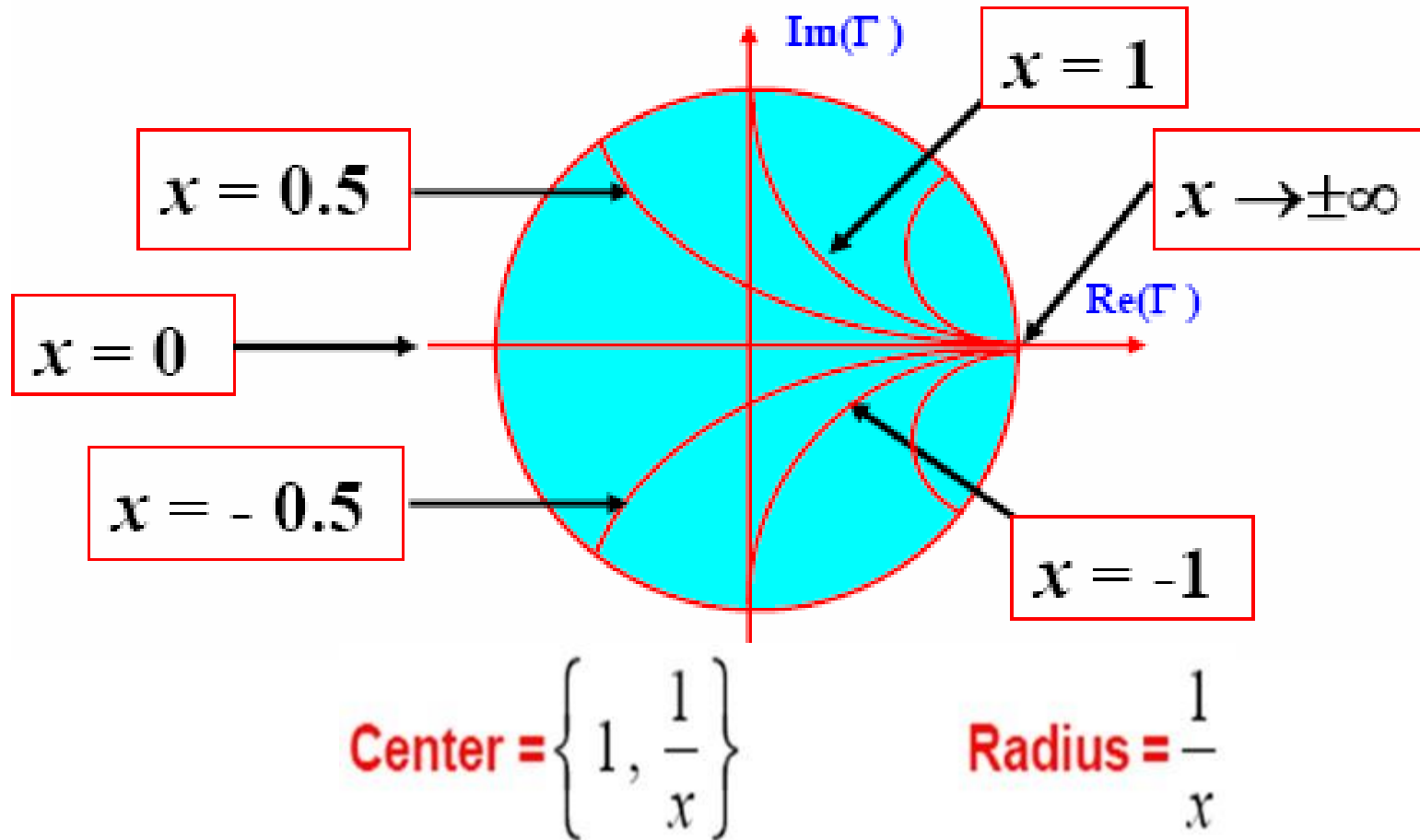
$$\left(\Gamma_r - \frac{r_L}{1 + r_L} \right)^2 + \Gamma_i^2 = \left(\frac{1}{1 + r_L} \right)^2,$$
$$(\Gamma_r - 1)^2 + \left(\Gamma_i - \frac{1}{x_L} \right)^2 = \left(\frac{1}{x_L} \right)^2$$

The result for the **imaginary part** indicates that on the complex plane with coordinates $(\text{Re}(\Gamma), \text{Im}(\Gamma))$ all the possible impedances with a given normalized reactance **x** are found on a **circle** with

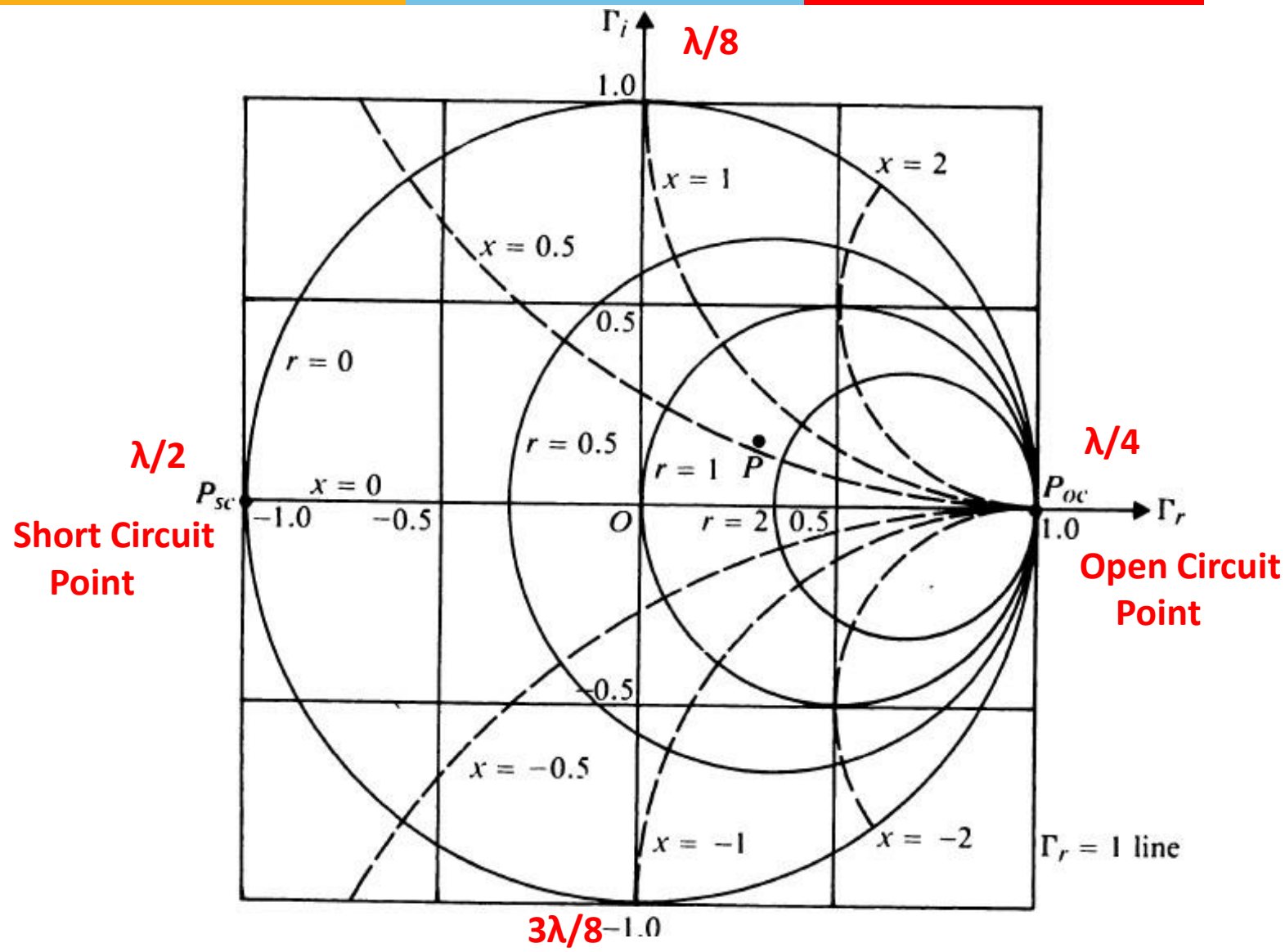
$$\text{Center} = \left\{ 1, \frac{1}{x} \right\} \quad \text{Radius} = \frac{1}{x}$$

As the normalized reactance **x** varies from $-\infty$ to ∞ , we obtain a family of arcs contained inside the domain of the reflection coefficient **$|\Gamma| \leq 1$** .

x-circles



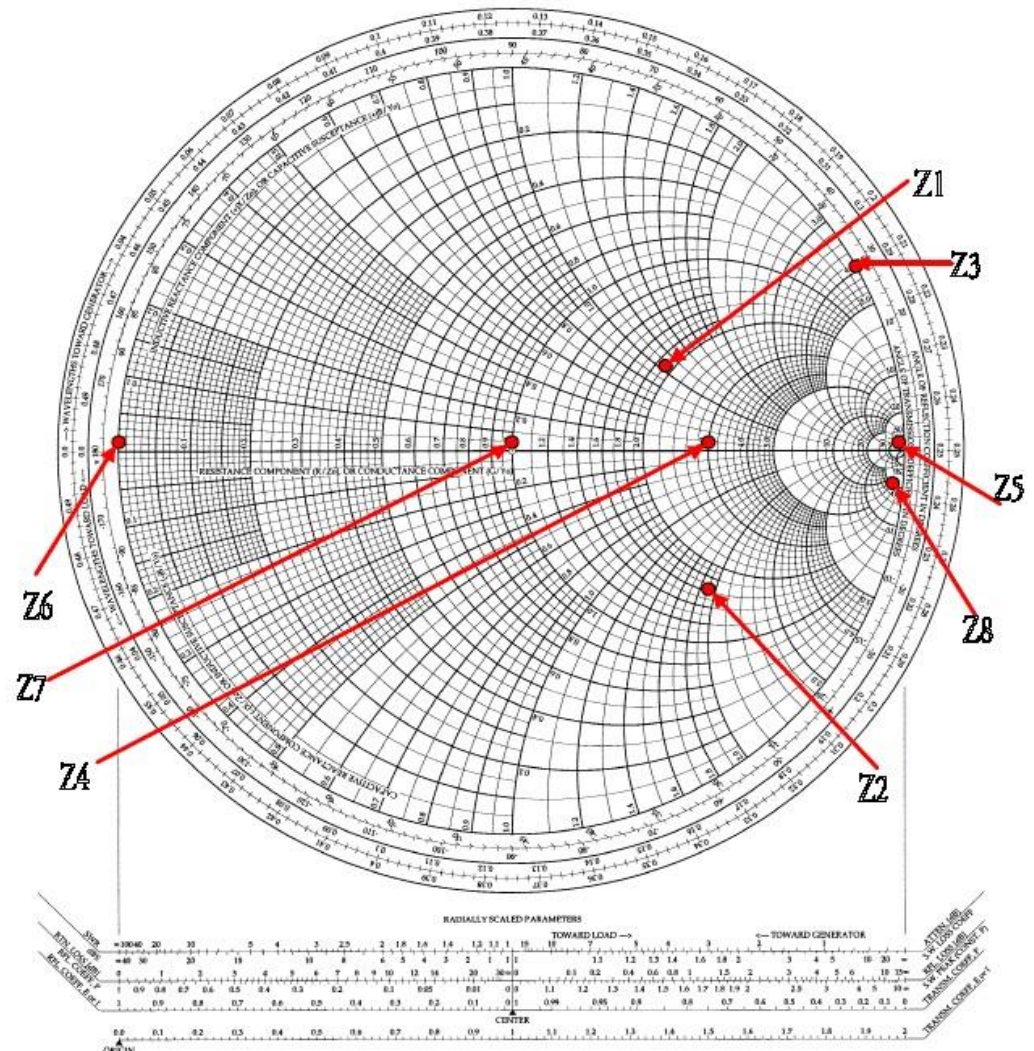
Smith Chart



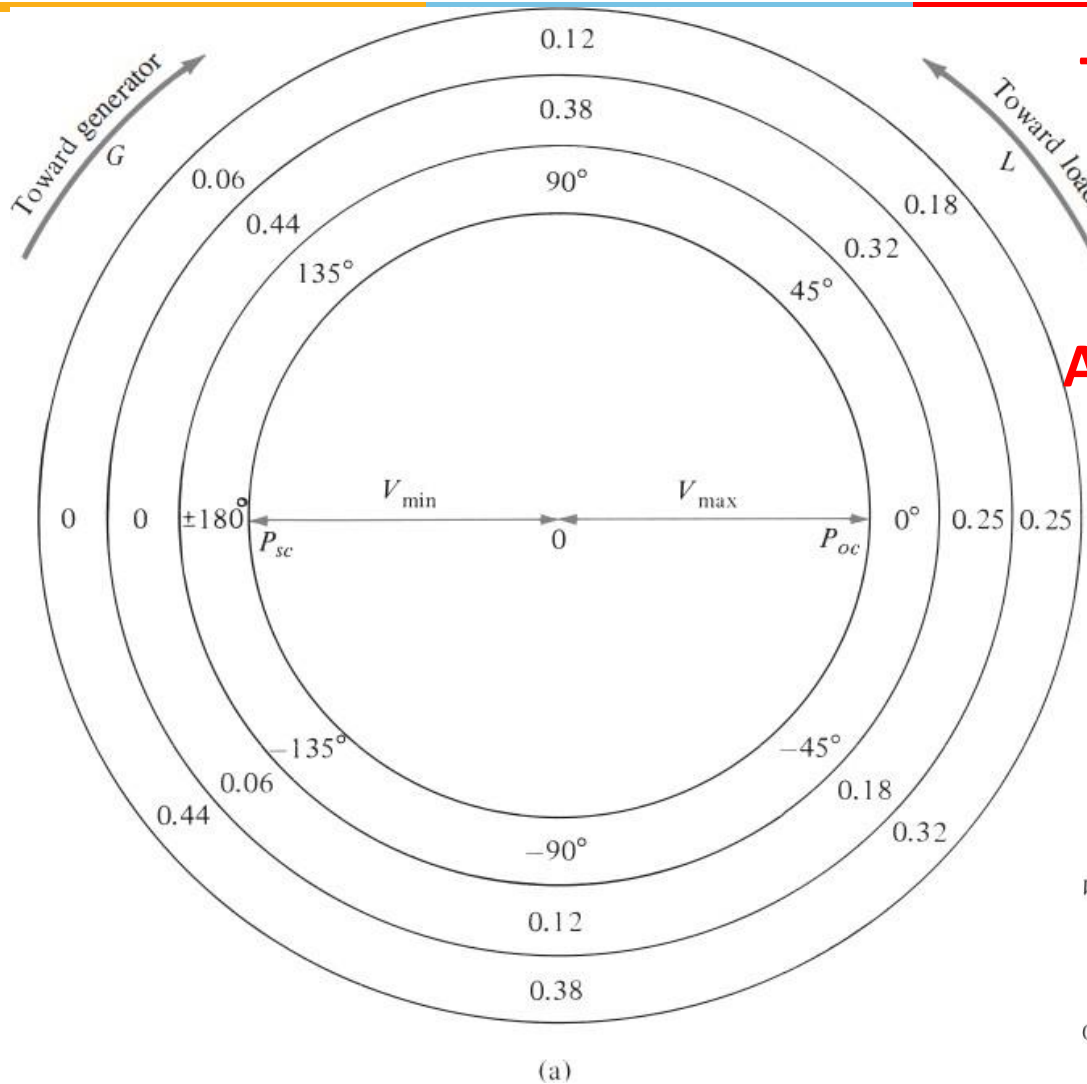
Important points about Smith chart



- Impedance divided by line impedance (50 Ohms)
 - $Z1 = 100 + j50$
 - $Z2 = 75 - j100$
 - $Z3 = j200$
 - $Z4 = 150$
 - $Z5 = \text{infinity}$ (an open circuit)
 - $Z6 = 0$ (a short circuit)
 - $Z7 = 50$
 - $Z8 = 184 - j900$
- Then, normalize and plot. The points are plotted as follows:
 - $z1 = 2 + j$
 - $z2 = 1.5 - j2$
 - $z3 = j4$
 - $z4 = 3$
 - $z5 = \text{infinity}$
 - $z6 = 0$
 - $z7 = 1$
 - $z8 = 3.68 - j18S$

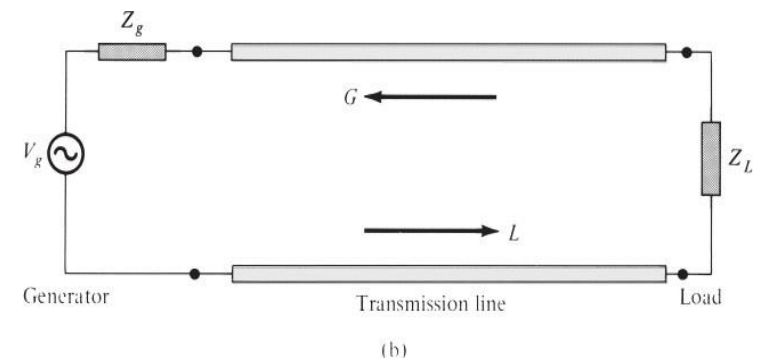


Important points about Smith chart



Towards Load: Anticlockwise

Away from Load: Clockwise

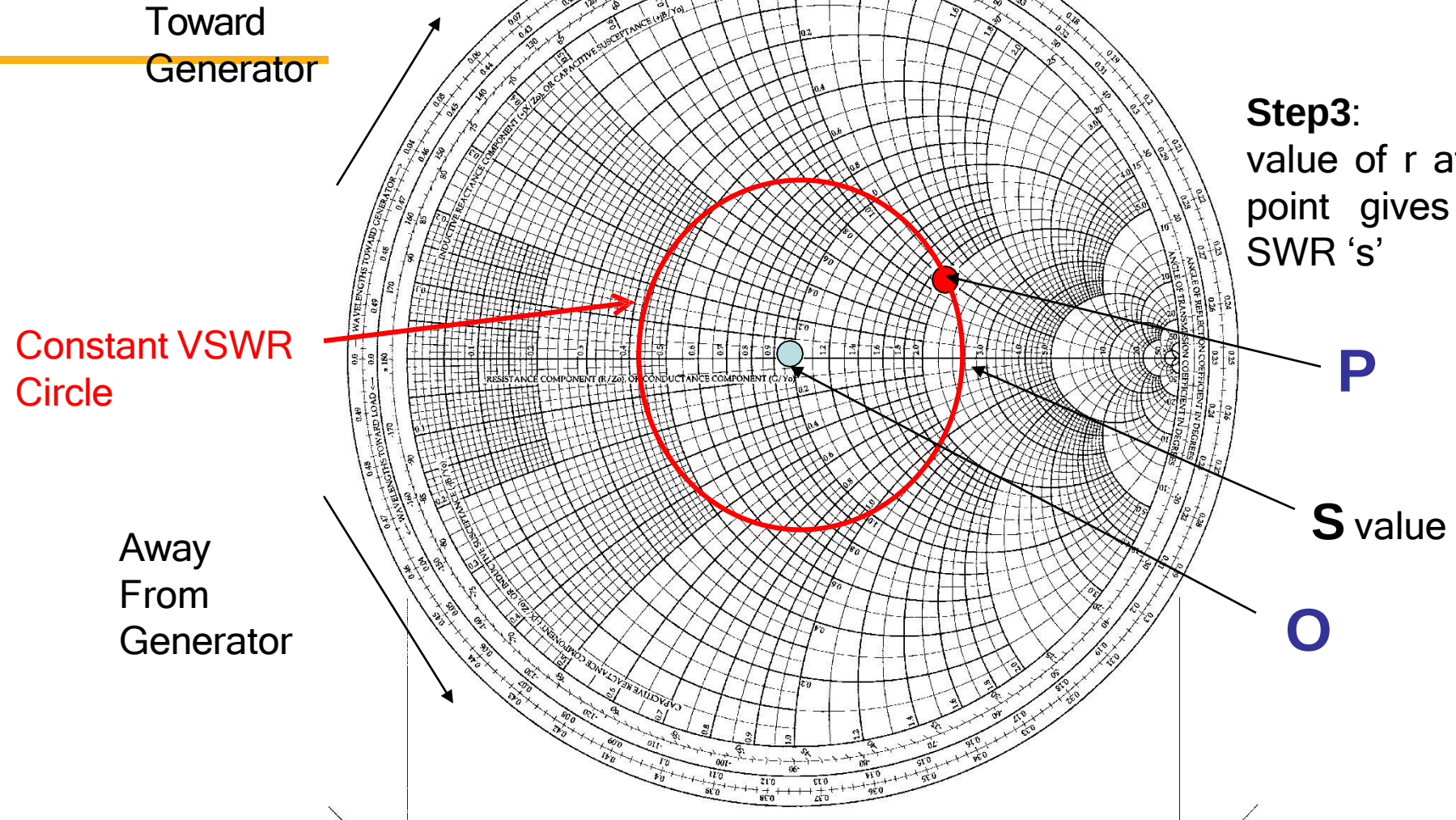


Plotting a constant VSWR circle

Step1: Locate the normalized load impedance on the chart (say P)

achieve lead

Step3: The value of r at this point gives you SWR 's'



Step2: Take center of the Smith chart (O) as the center and the radius as OP and draw a circle. (The circle passes through the normalized load impedance).

Learning Example



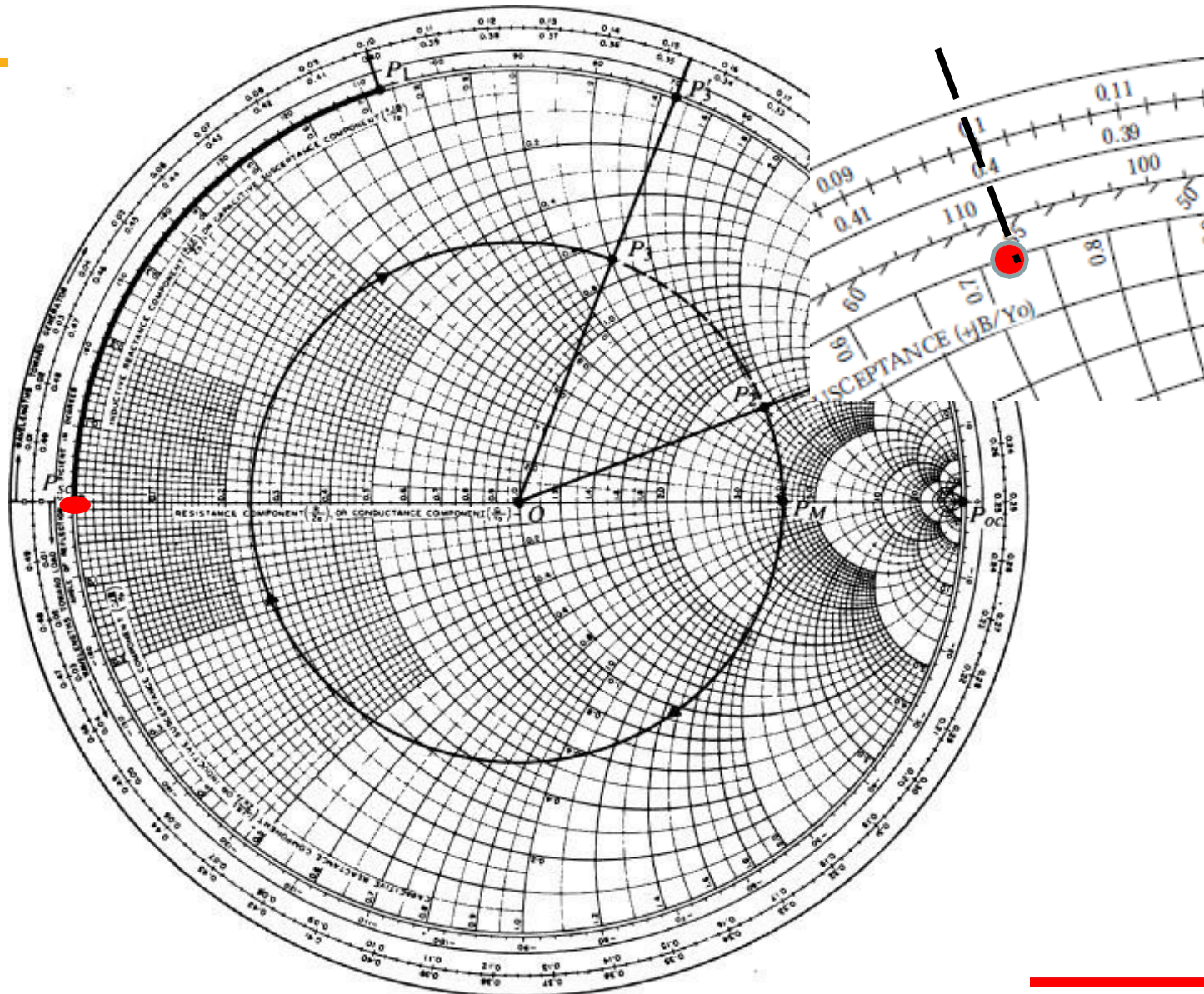
Find the input impedance of a section of a $50\ \Omega$ loss-less transmission line that is 0.1λ long and is terminated in a short circuit.

Learning Example

innovate

achieve

lead



To find the reflection coefficient for a known load

1. Normalization

$$\begin{aligned} z_n &= (25 + j\ 100)/50 \\ &= 0.5 + j\ 2.0 \end{aligned}$$

3. Find normalized reactance arc

$$x = 2.0$$

2. Find normalized resistance circle

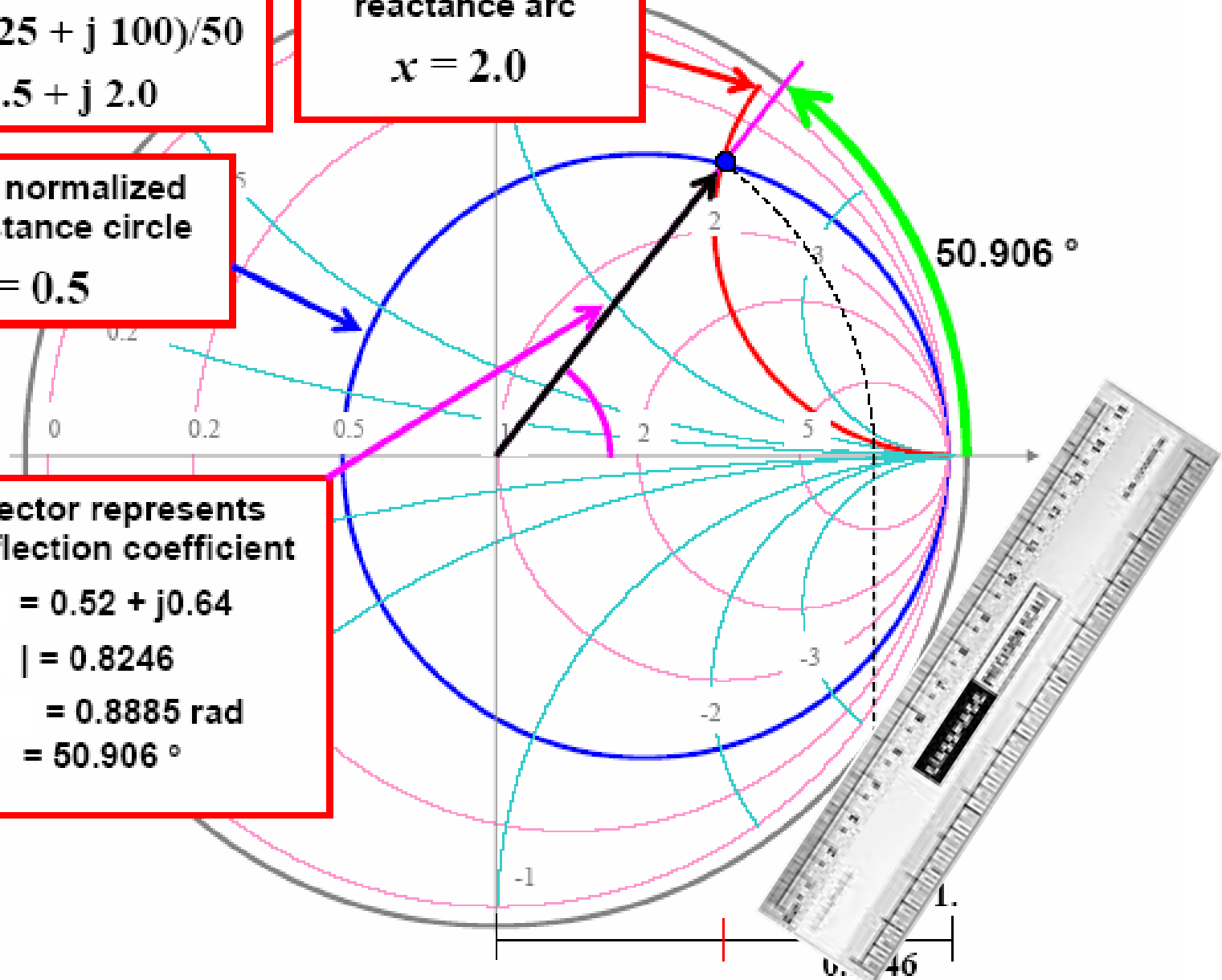
$$r = 0.5$$

4. This vector represents the reflection coefficient

$$\Gamma = 0.52 + j0.64$$

$$|\Gamma| = 0.8246$$

$$\begin{aligned} \angle \Gamma &= 0.8885 \text{ rad} \\ &= 50.906^\circ \end{aligned}$$



Example

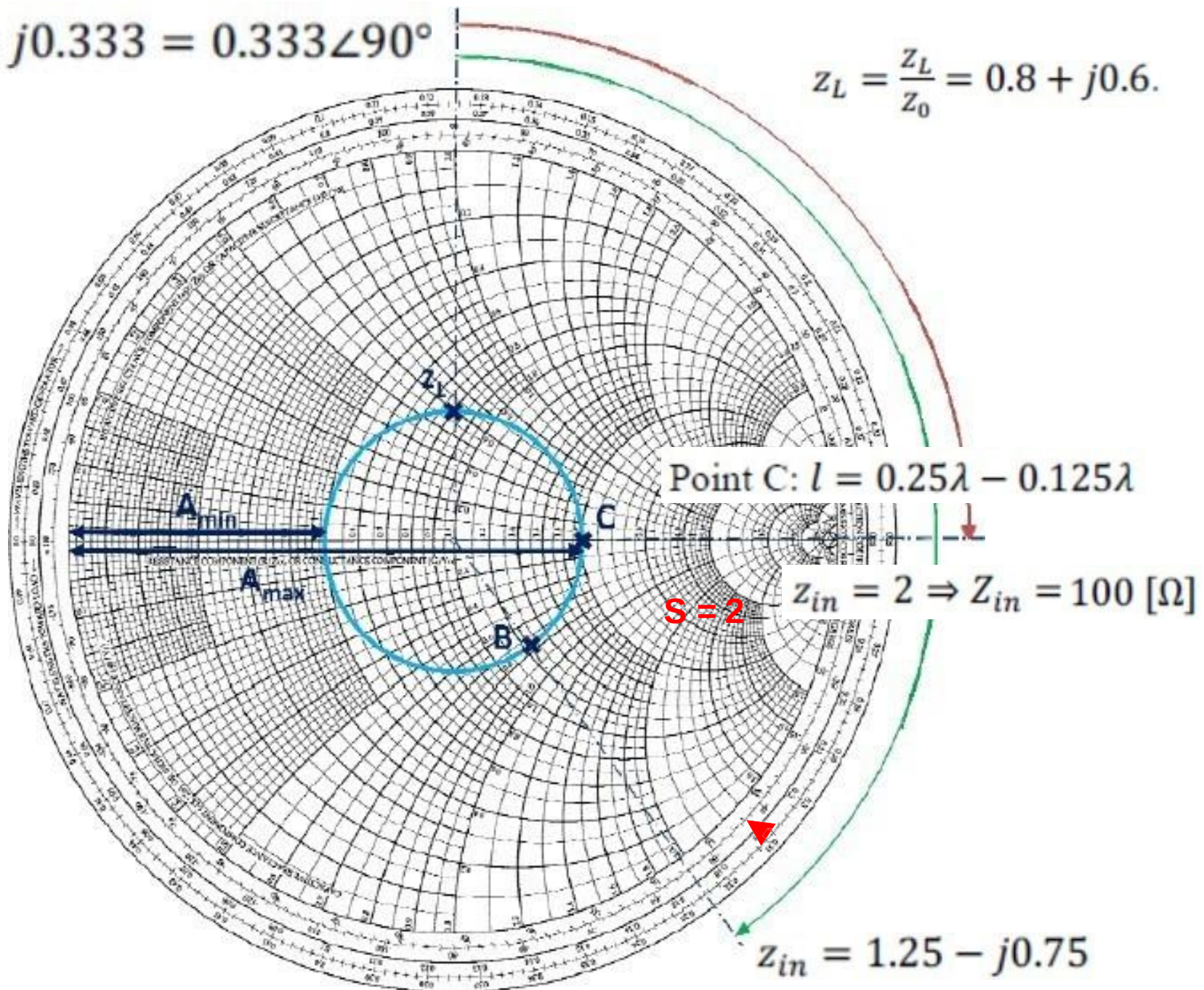


A lossless transmission line with a characteristic impedance of 50Ω is terminated in a load impedance $40 + j30 \Omega$. Using Smith Chart techniques, determine:

- The reflection coefficient Γ (magnitude and phase), and the VSWR
- The input impedance seen looking into the line if it is 0.2λ long.
- The length of line needed to make the input impedance look real (one solution is sufficient), and the associated resistance value.

a) $\Gamma = j0.333 = 0.333\angle 90^\circ$

$$z_L = \frac{Z_L}{Z_0} = 0.8 + j0.6.$$



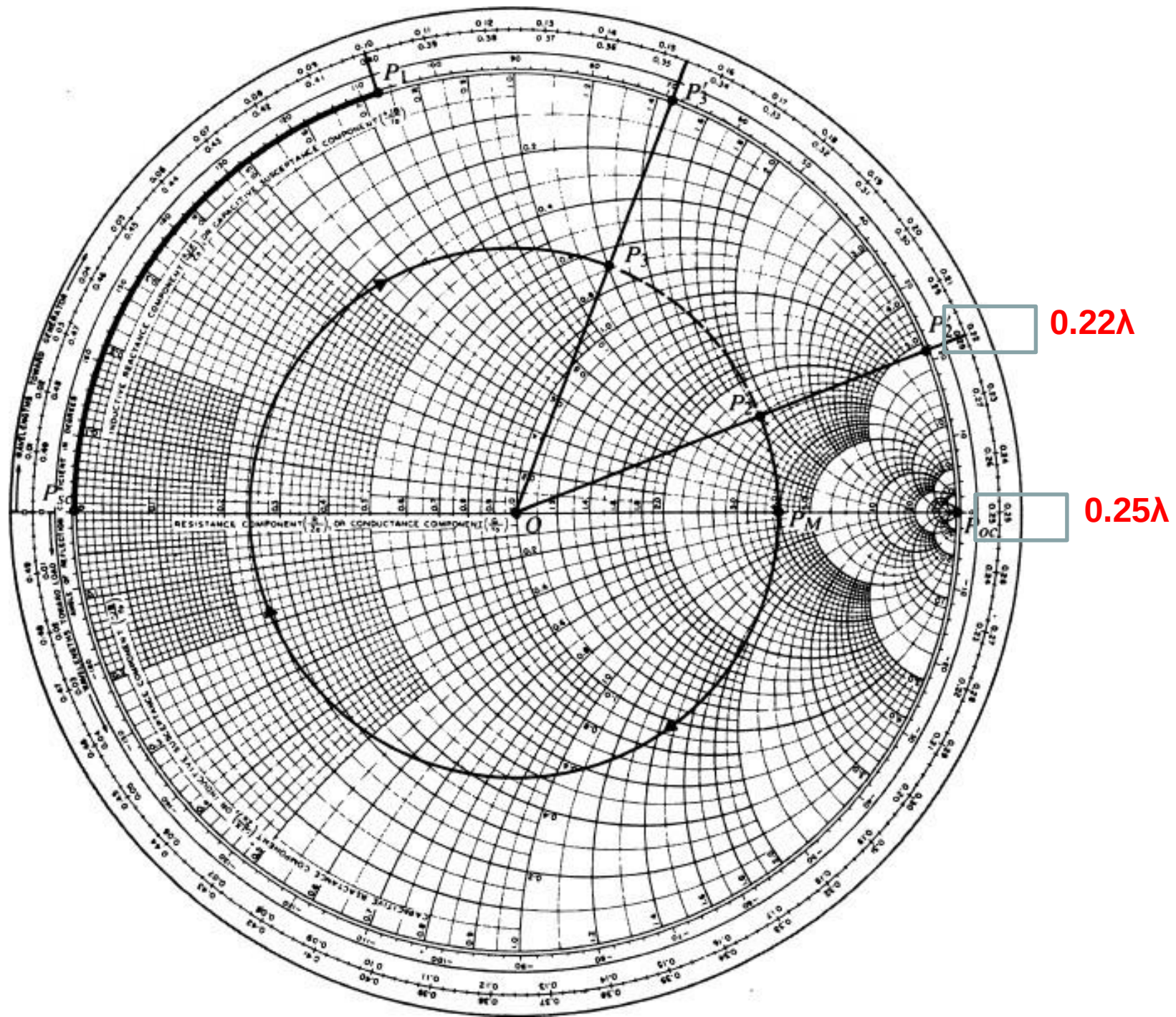
Example

EXAMPLE 9–14 A lossless transmission line of length 0.434λ and characteristic impedance $100\ \Omega$ is terminated in an impedance $260 + j180\ \Omega$. Find (a) the voltage reflection coefficient, (b) the standing-wave ratio, (c) the input impedance, and (d) the location of a voltage maximum on the line.

$$z' = 0.434\lambda,$$

$$R_0 = 100\ \Omega,$$

$$Z_L = 260 + j180$$



Why Impedance Matching



- **Malfunctioning of the source**
- **Breakdown (High VSWR)**
- **Frequency Pulling**

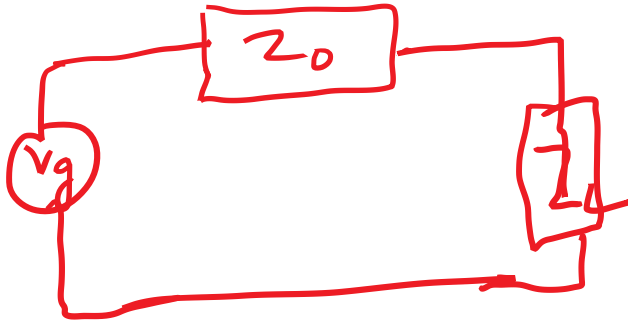
Sol: Impedance Matching

Impedance Matching

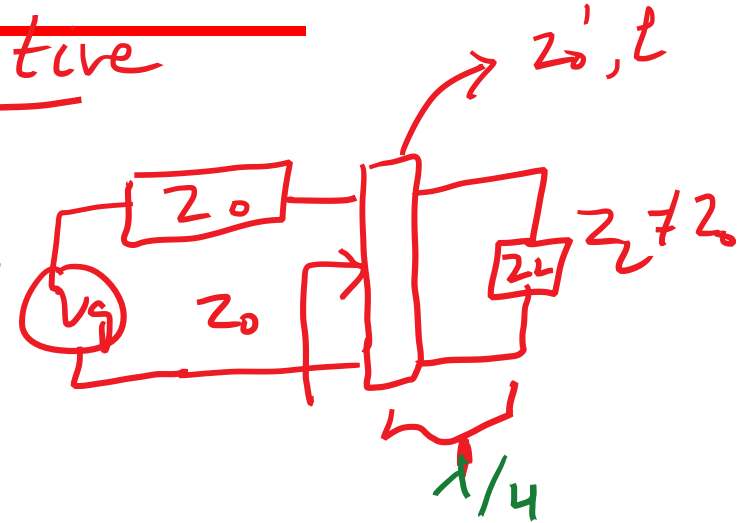


lossless

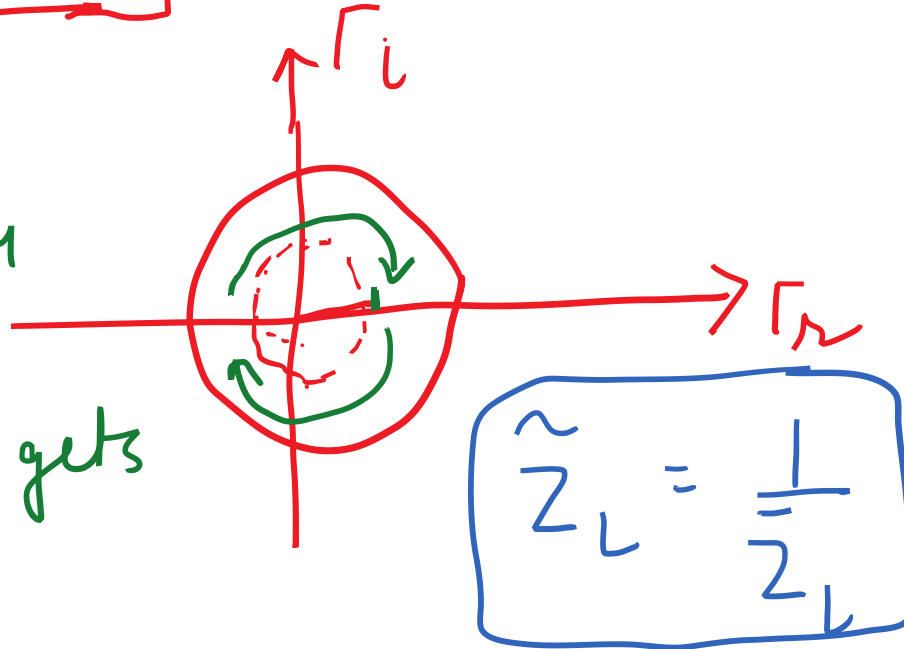
Z_L is resistive



$$Z_0 \neq Z_L \rightarrow$$



At every $\lambda/4$
Normalized
impedance
inverted
gets



Resistive
to
Resistive
matching

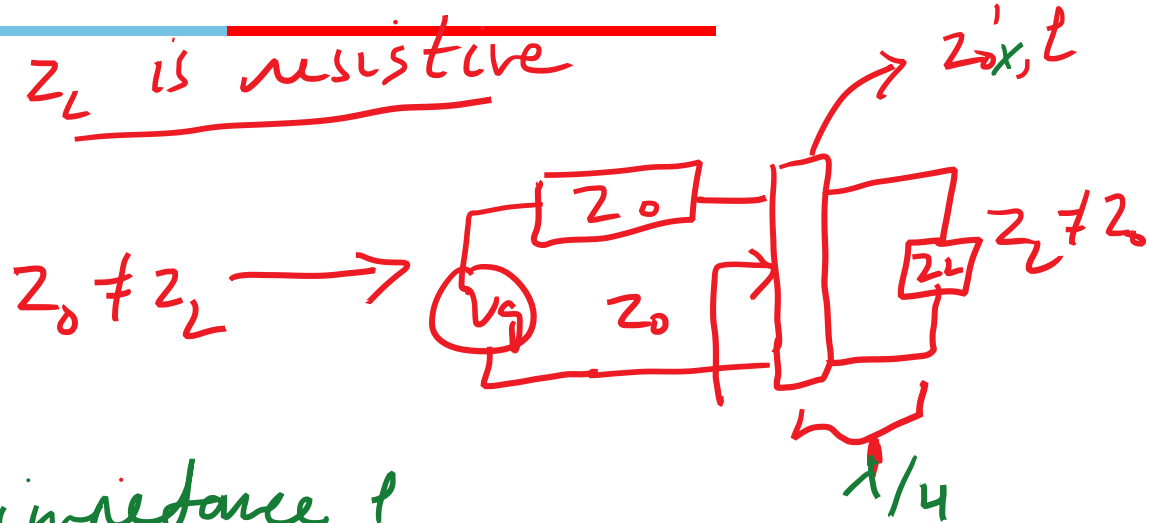
Impedance Matching



Quarter Wave Transformer

lossless

Z_L is resistive



$l = \lambda/4$

Normalized load impedance Γ

$\frac{Z_L}{Z_0}$ after $\lambda/4$ $\left(\frac{Z_0}{Z_L} \right)$ Normalized Resistive value to

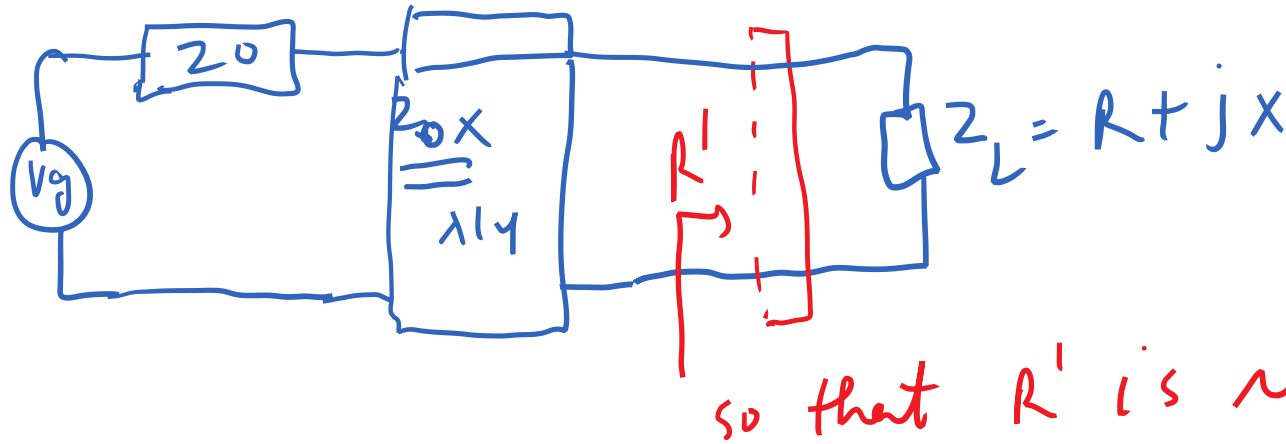
So Actual impedance $\rightarrow \frac{Z_0^2}{Z_L}$ Resistive matching

For matching $\rightarrow \left[Z_0 = \frac{Z_0^2}{Z_L} \right]$ so $\left[Z_0 = \sqrt{Z_0 Z_L} \right]$

Quarter Wave Transformer



What about complex impedances??



so that R' is resistive

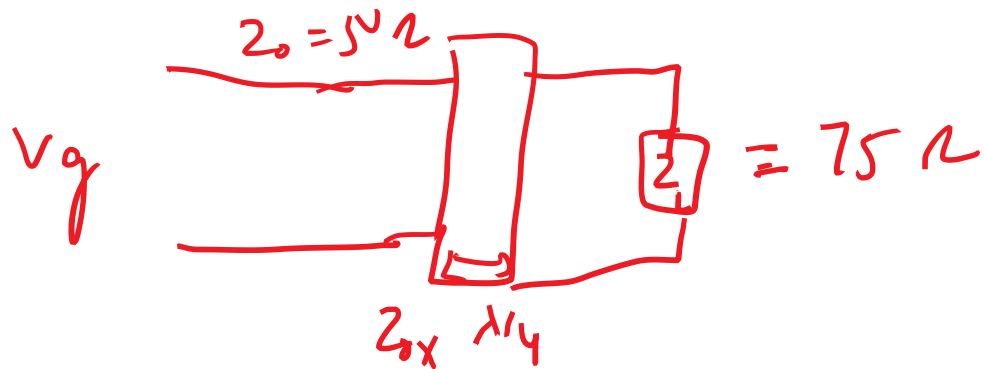
Is sweet ✓ $Z_o X$ for every value not available
✓ single frequency use

Example



Match the following load impedance to
 $Z_0 = 50 \Omega$ line

a) $Z_L = 75 \Omega$ b) $Z_L = 100 + j75 \Omega$

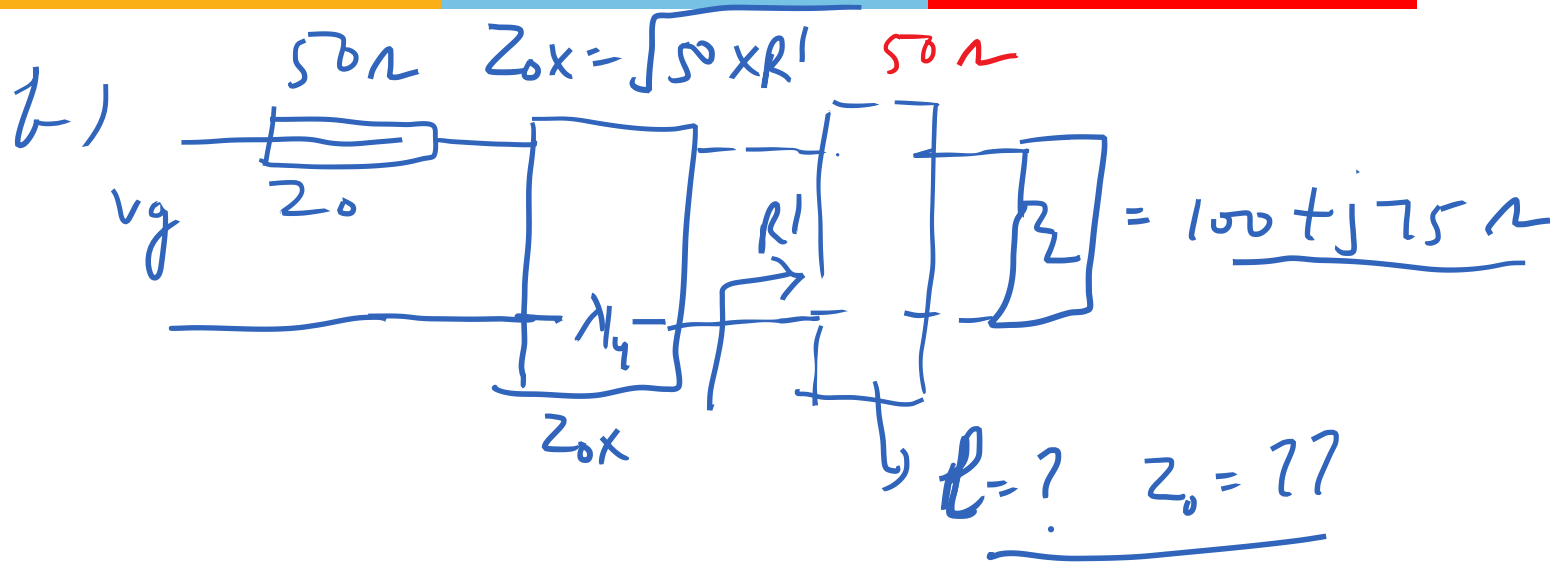


$$\text{Sol}^n \quad Z_{0x} = \sqrt{75 \times 50}$$

at $\frac{\lambda}{4}$

$$Z_{0x} = 61.23 \Omega$$

Example

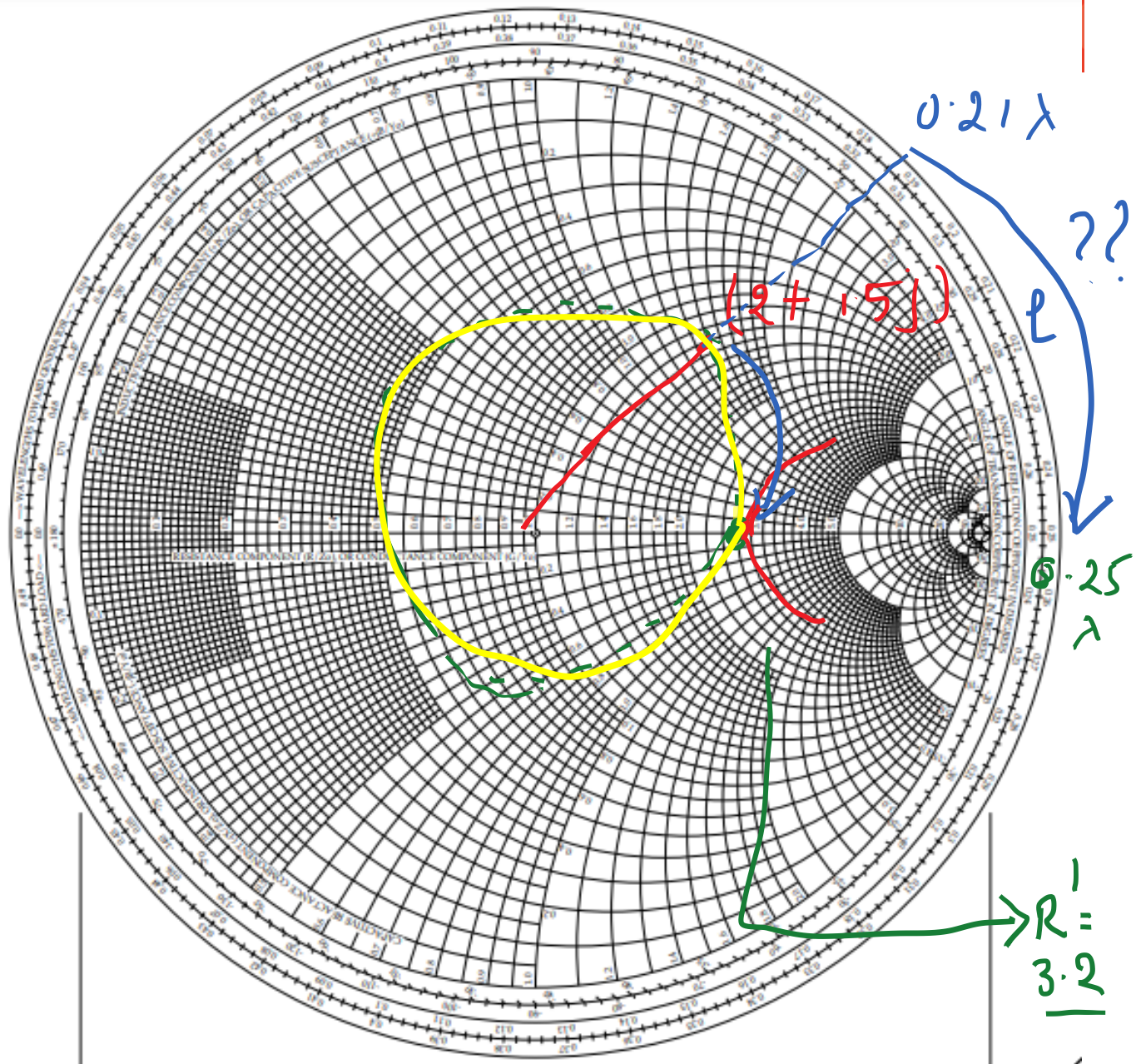


$$Z_L' = 2 + 1.5j$$

→ Draw on Smith chart and

find the value of l at which impedance is real.

Example

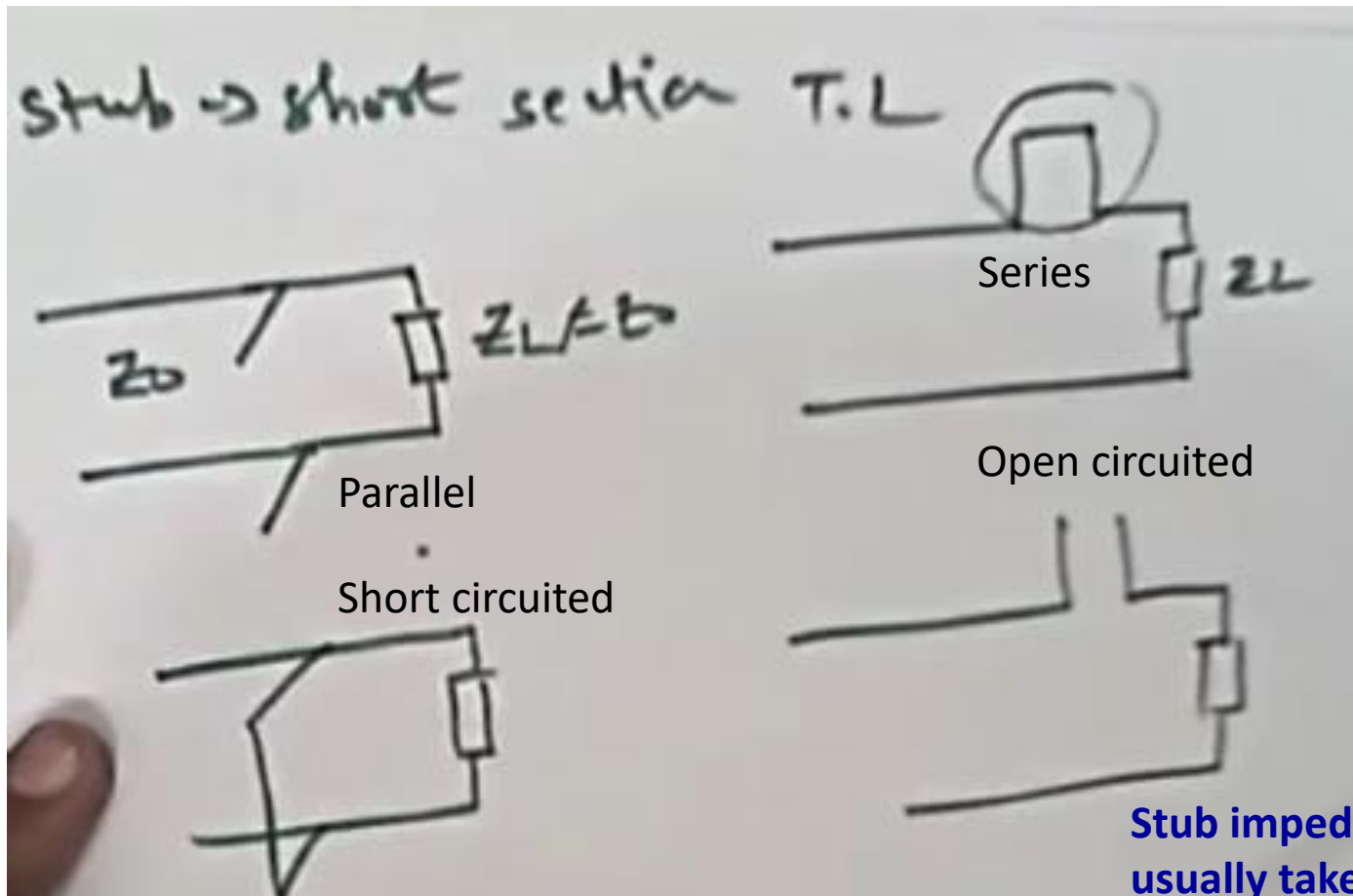


Impedance Matching with stub



Stub : A short section of transmission line

- Single Stub
- Double Stub



Stub impedance is usually taken as characteristic impedance of transmission line

To find the admittance for a known load

$$Z_L = 30 + j70$$



Step1:

Normalized impedance

step2

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$$\Gamma_L = \frac{\tilde{Z}_L - 1}{\tilde{Z}_L + 1}$$

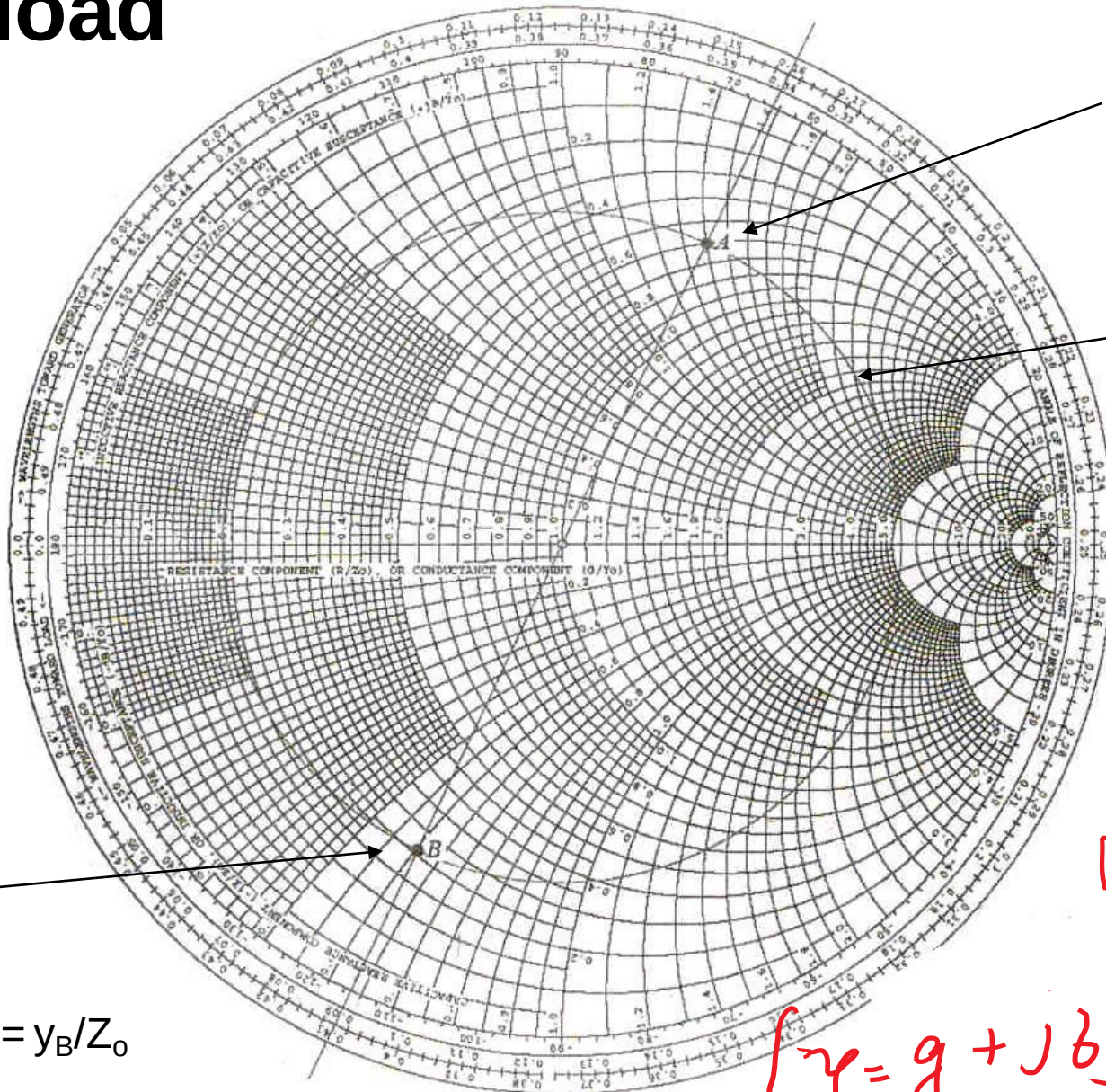
$$\begin{aligned} \Gamma_L &= \frac{Y_0 - Y_L}{Y_0 + Y_L} \\ &= \frac{1 - \tilde{Y}_L}{1 + \tilde{Y}_L} \end{aligned}$$

$$\Gamma_L = -\frac{(\tilde{Y}_L - 1)}{(\tilde{Y}_L + 1)}$$

$$[\gamma = g + jb]$$

Y_B

$$Y = Y_0 Y_B = Y_B / Z_0$$

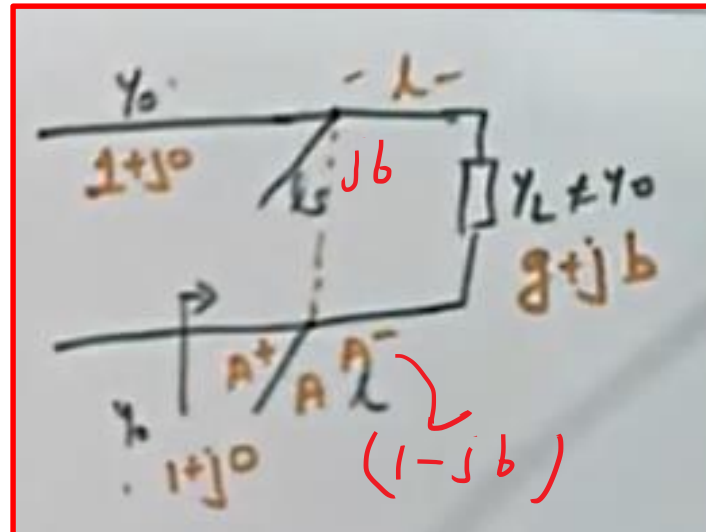
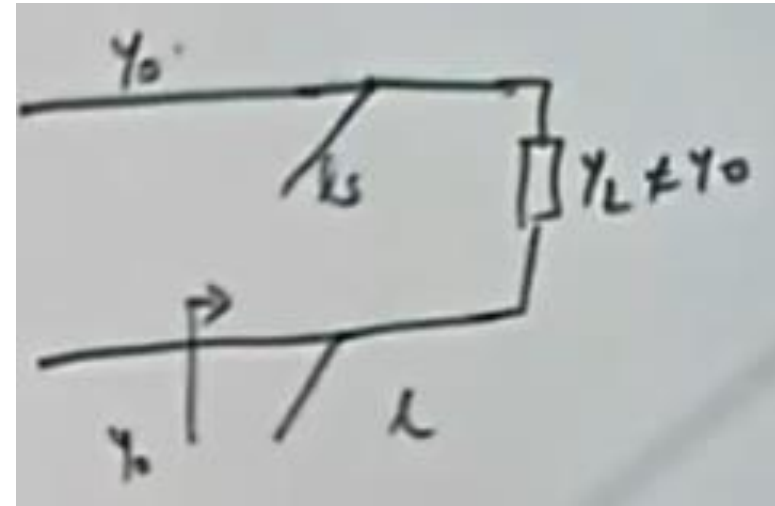
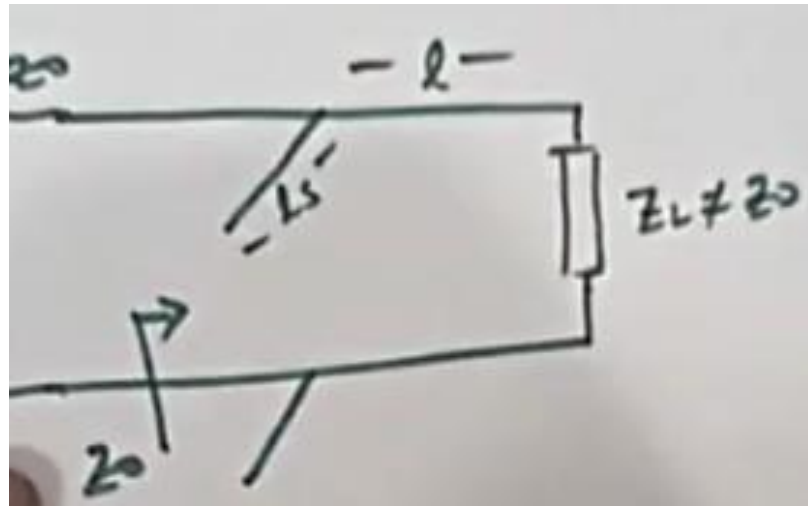


Parallel Single stub (open circuited)

innovate

achieve

lead



$$Z_{oc} = -j Z_0 \cot \beta L$$

$$Y_{oc} = \frac{j \tan \beta L}{Z_0}$$

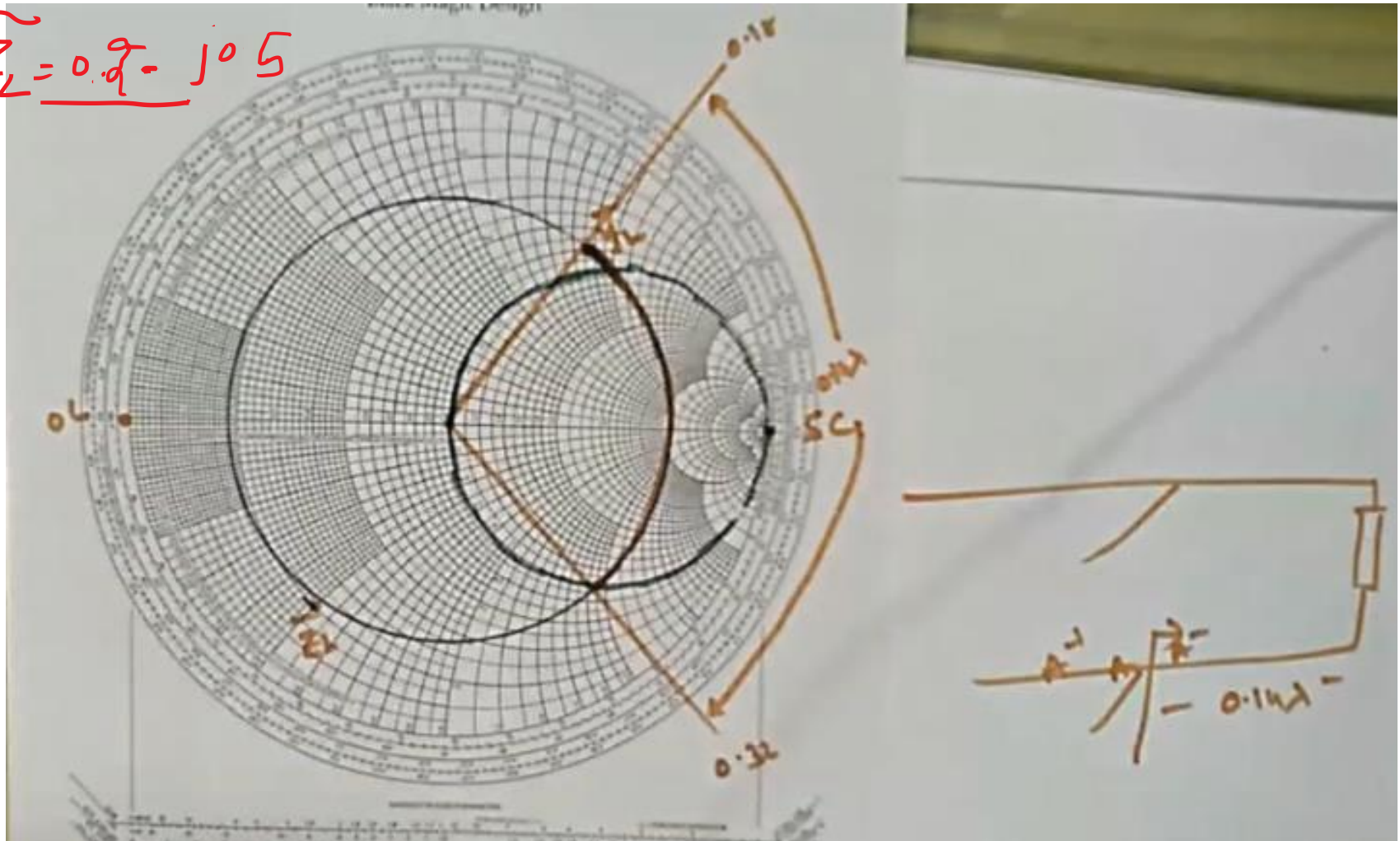
$$Z_{sc} = j Z_0 \tan \beta L$$

$$Y_{sc} = \frac{-j \cot \beta L}{Z_0}$$

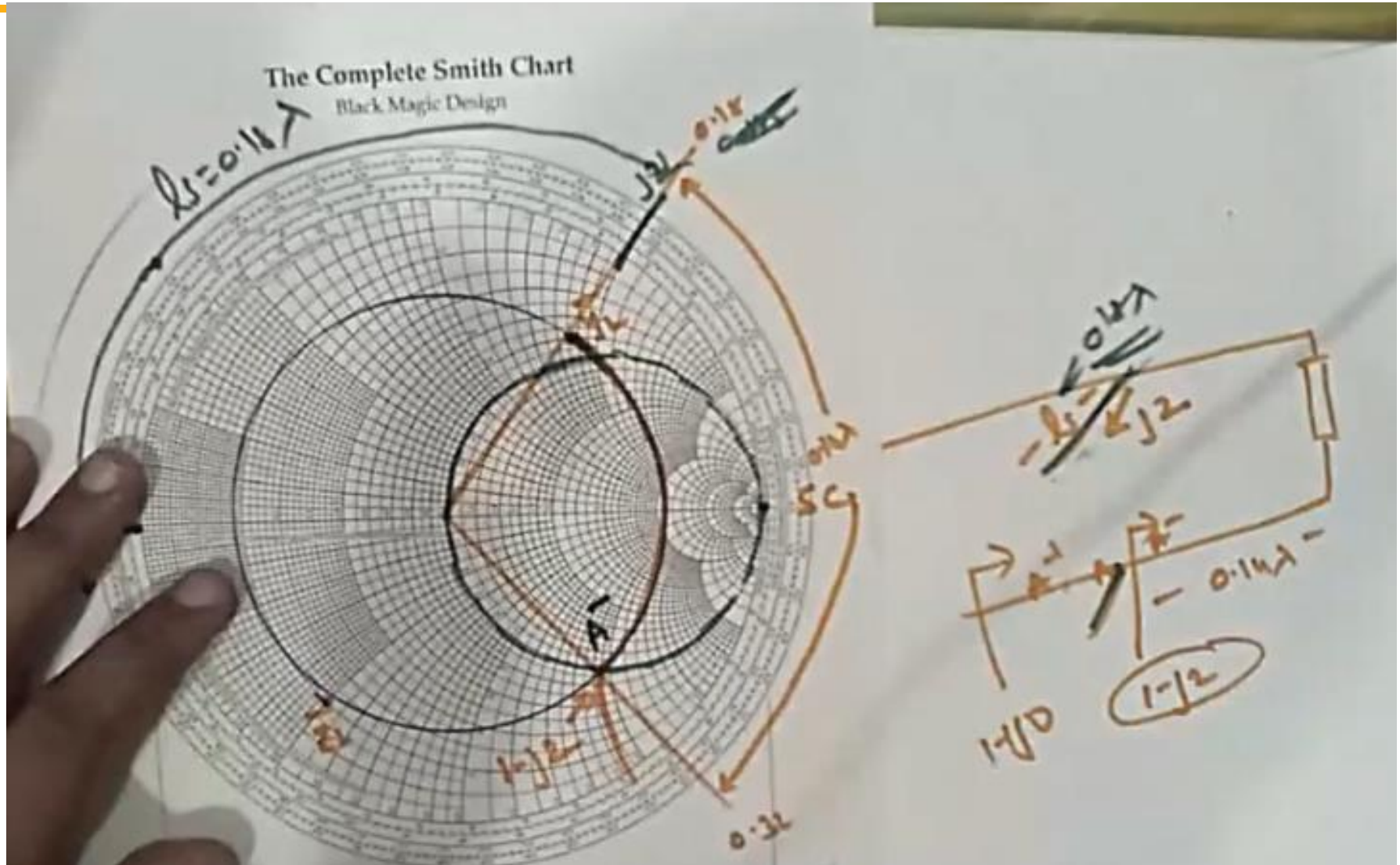
Example



$$\tilde{Z}_L = 0.2 - j0.5$$



Example



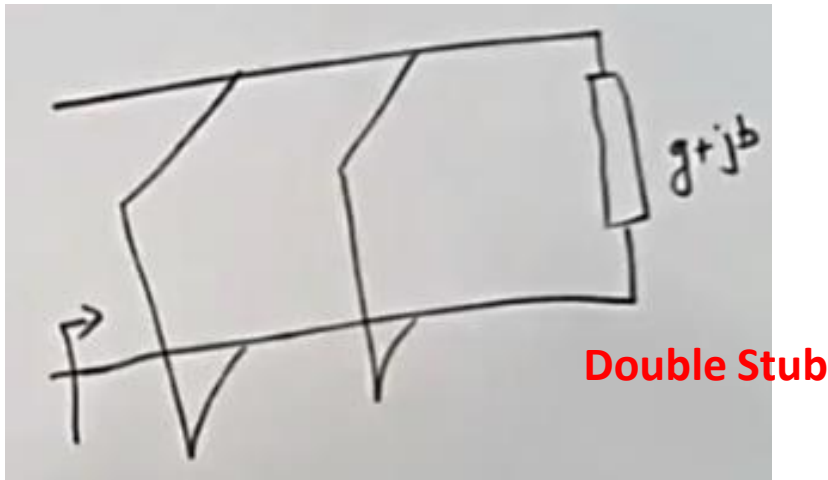
Parallel Double stub (Short circuited)

innovate

achieve

lead

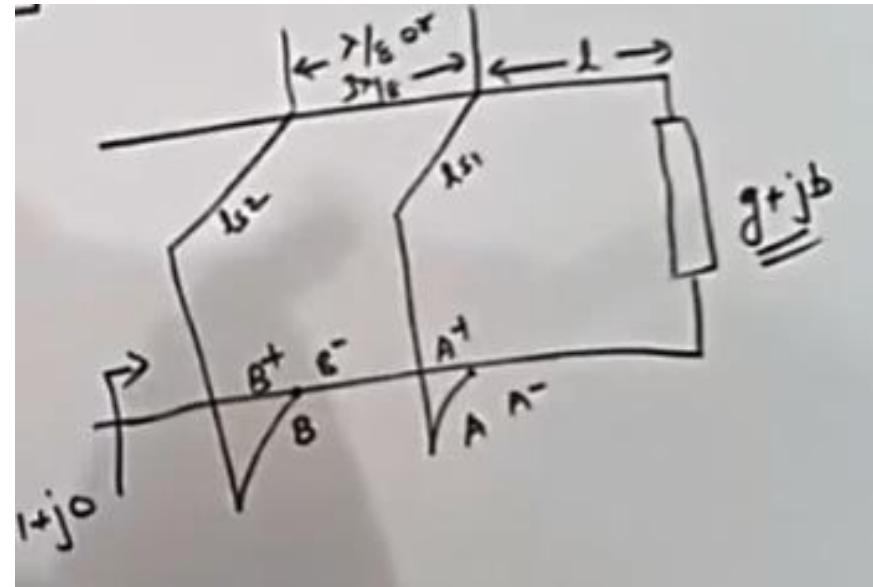
What if the Load is continuously changing?



$$\frac{3\lambda}{8}$$

$$2\beta l = 2 \times 2\pi \times \frac{3\lambda}{8} \times \frac{1}{2}$$

$$= 275^\circ$$

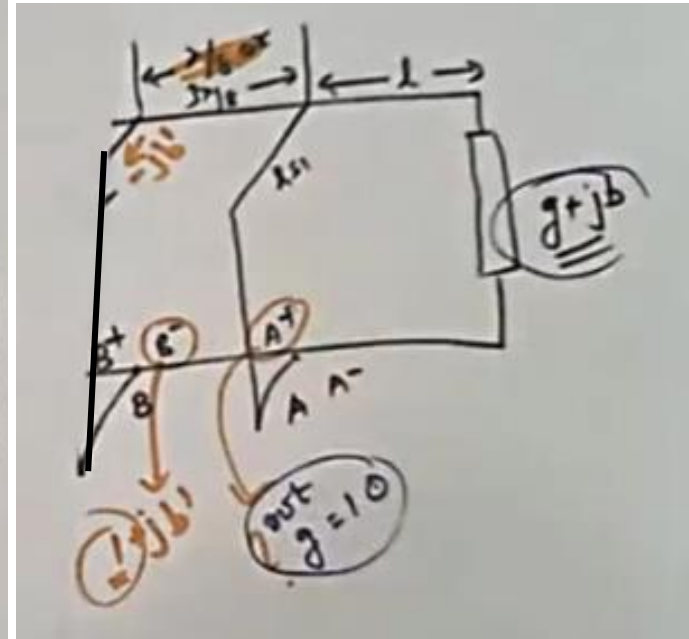
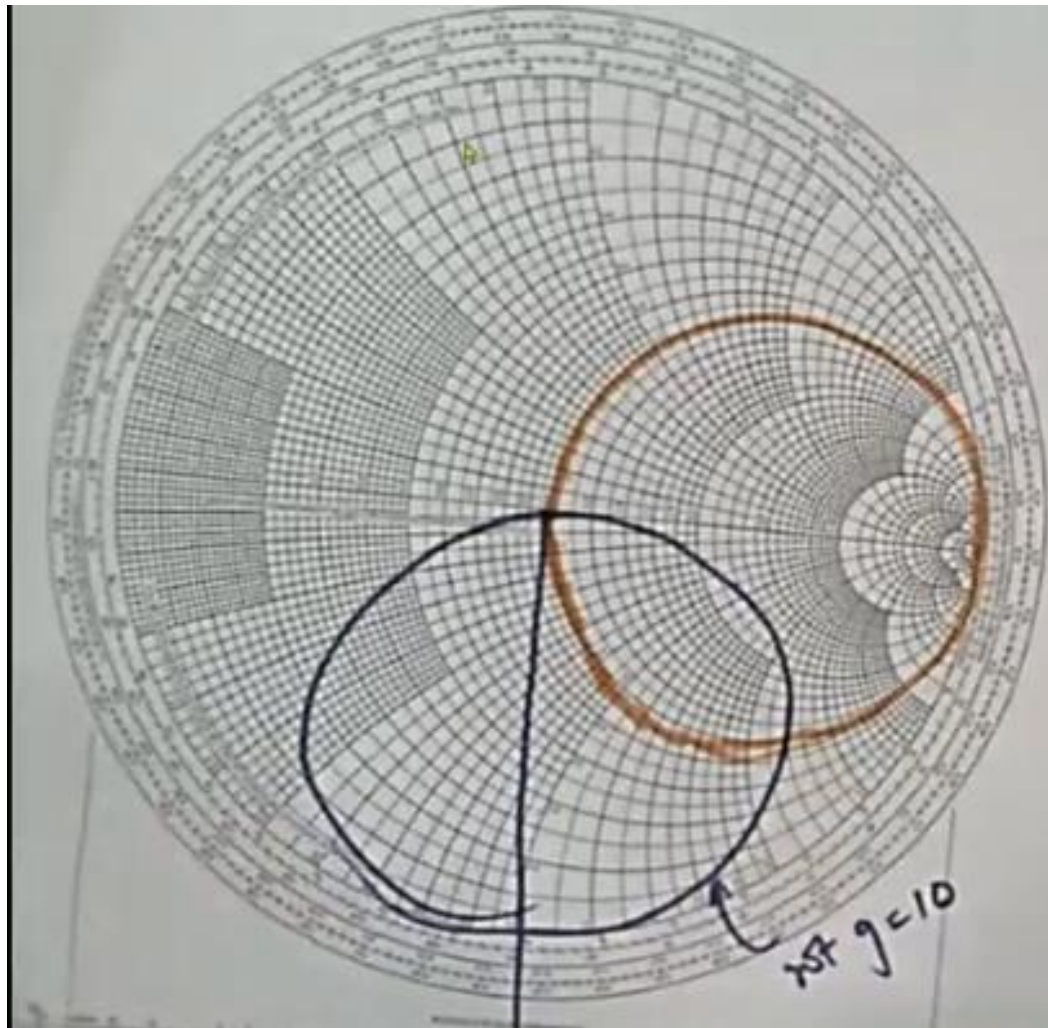


Parallel Double stub (Short circuited)

innovate

achieve

lead

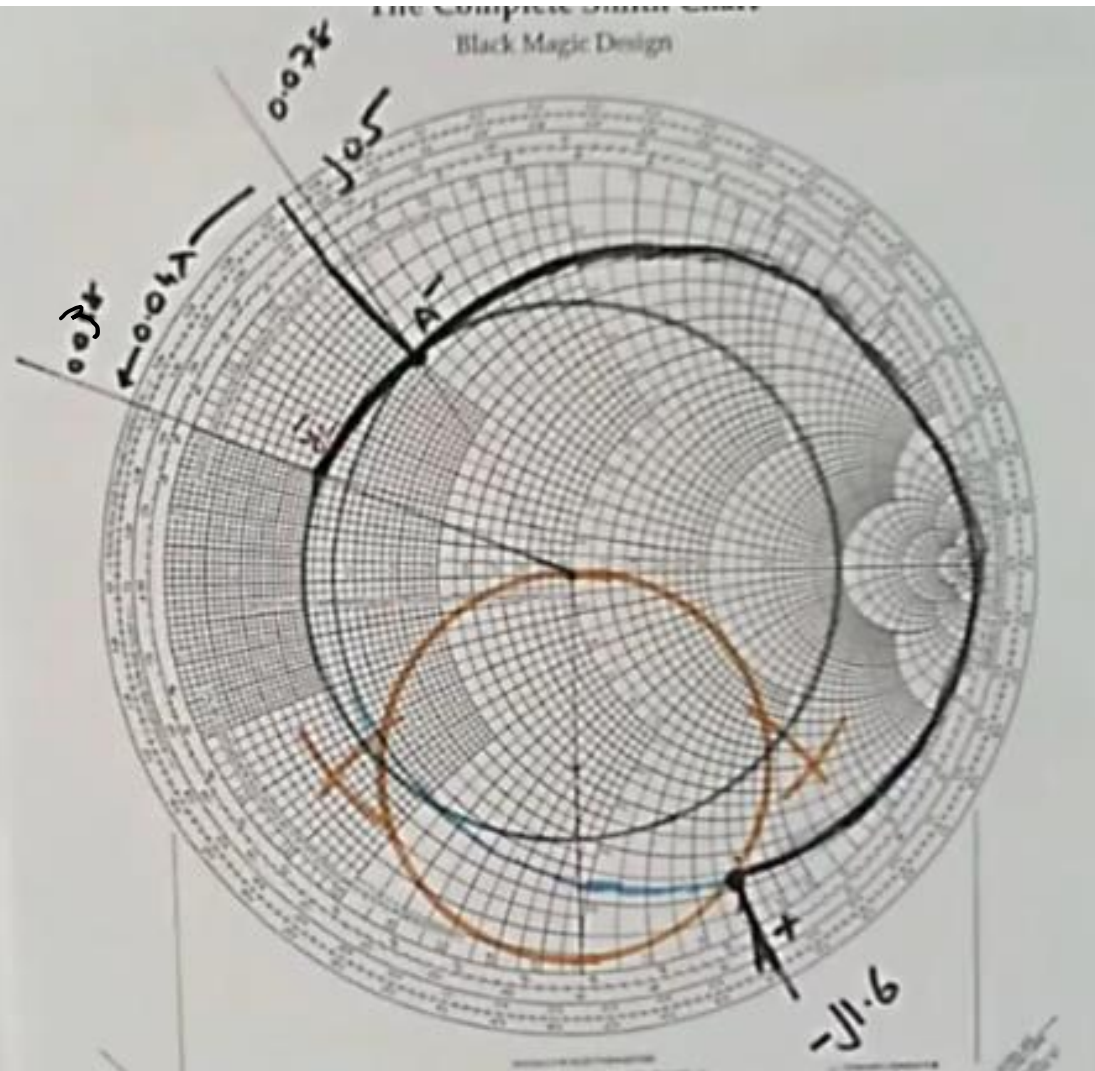


Parallel Double stub (Short circuited)

innovate

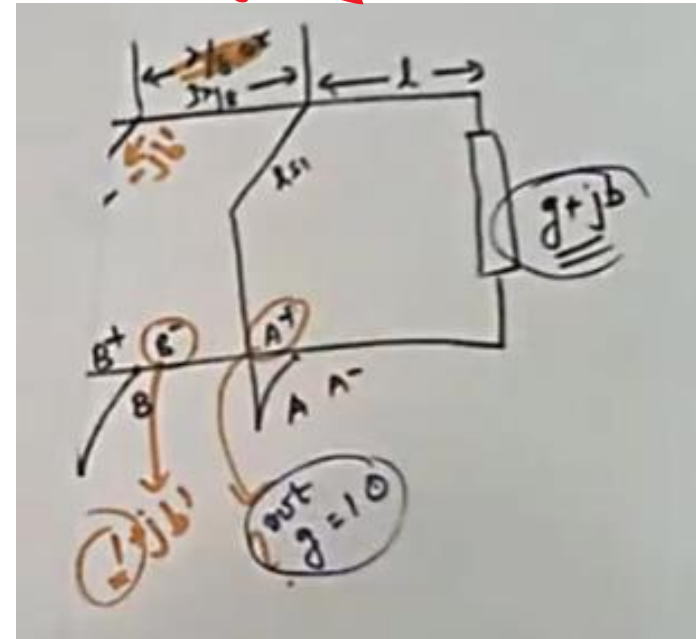
achieve

lead



$$\tilde{Z}_L = 2.5 - j2.5$$

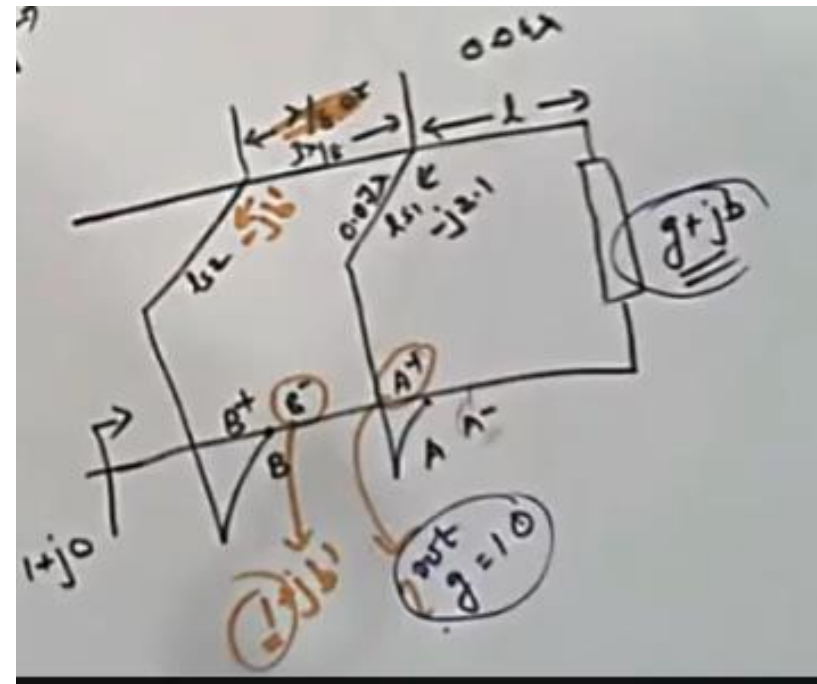
$$\tilde{Y}_L = 0.2 + j0.2$$



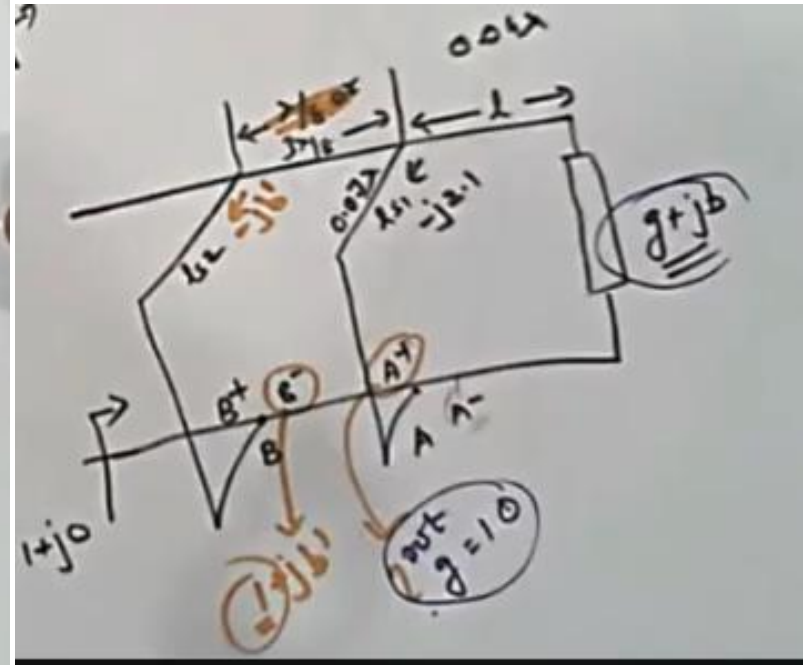
$$j0.5 + \text{stub} = -j1.6$$

$$\text{stub} = -2.1 \checkmark$$

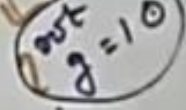
circuited)
innovate achieve lead



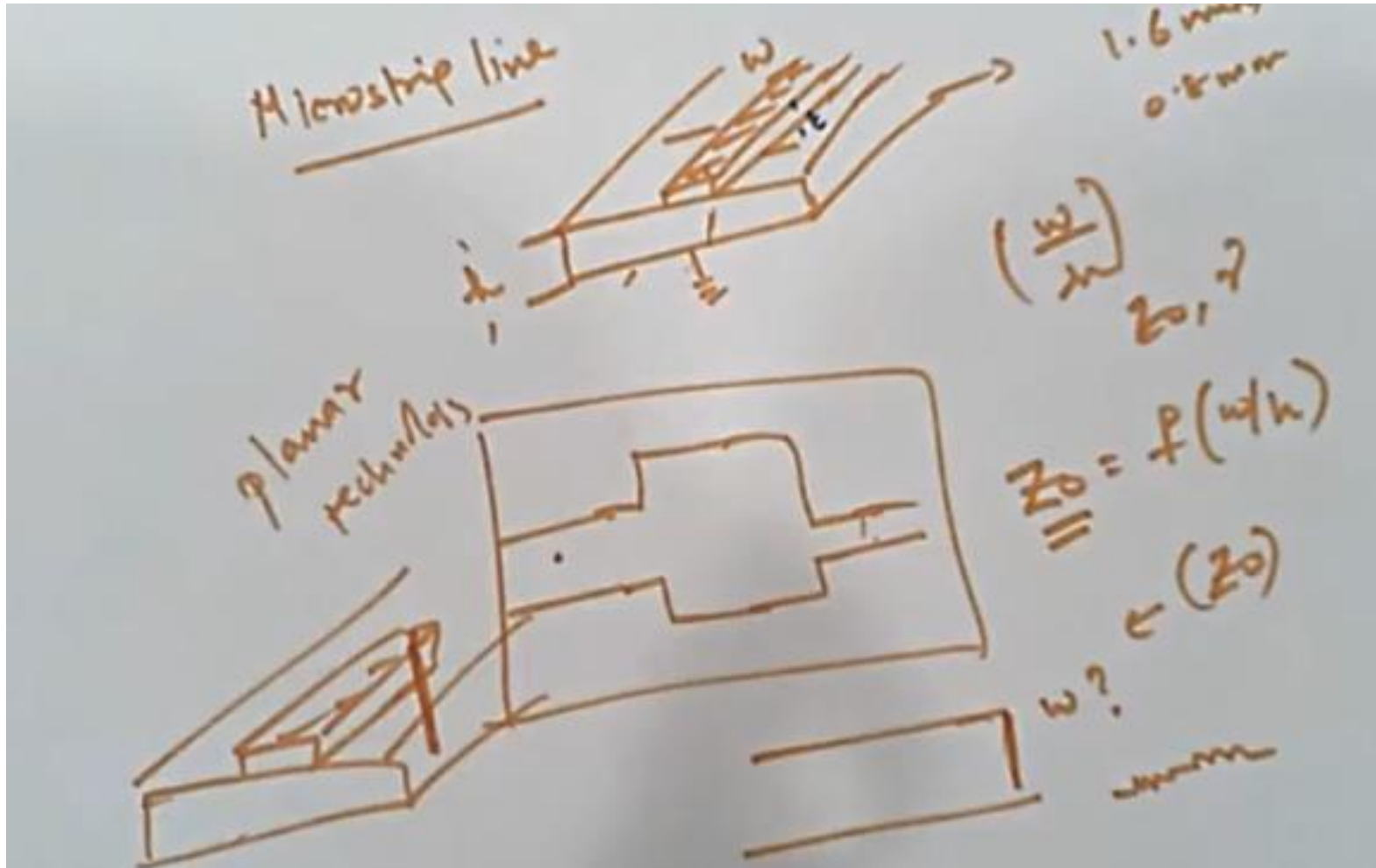
lead



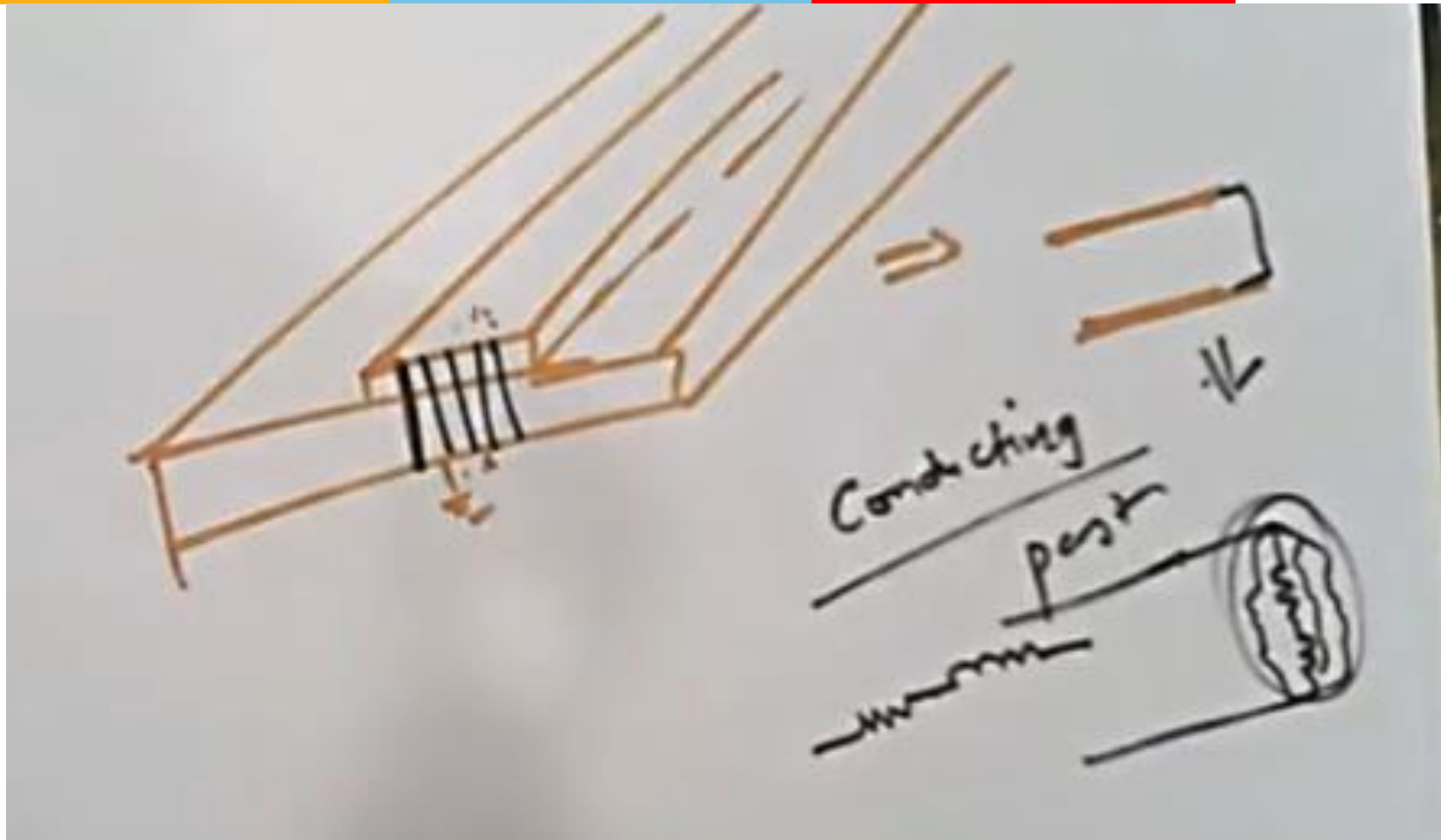
lead



Microstrip Line (Planar)



Microstrip Line



Power Relations



Power relations

Z_s , Z_0 , Z_L , Z_{in} , Γ

Z_s , Z_L

$$P = \text{Re}\{VI^*\}$$

peak values

$$V = \frac{V_{in}^+}{2} + \frac{V_{in}^-}{2} = \frac{V_{in}^+}{2}(1 + \Gamma_{in})$$

$$I = \frac{V_{in}^+}{Z_0} - \frac{V_{in}^-}{Z_0}$$

$$\Gamma = \frac{V_{in}^-}{V_{in}^+} = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$$P = \frac{1}{2} \left[\frac{(V_{in}^+)^2}{Z_0} (1 - |\Gamma_{in}|^2) \right] = P_{av}$$